

BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

Complete

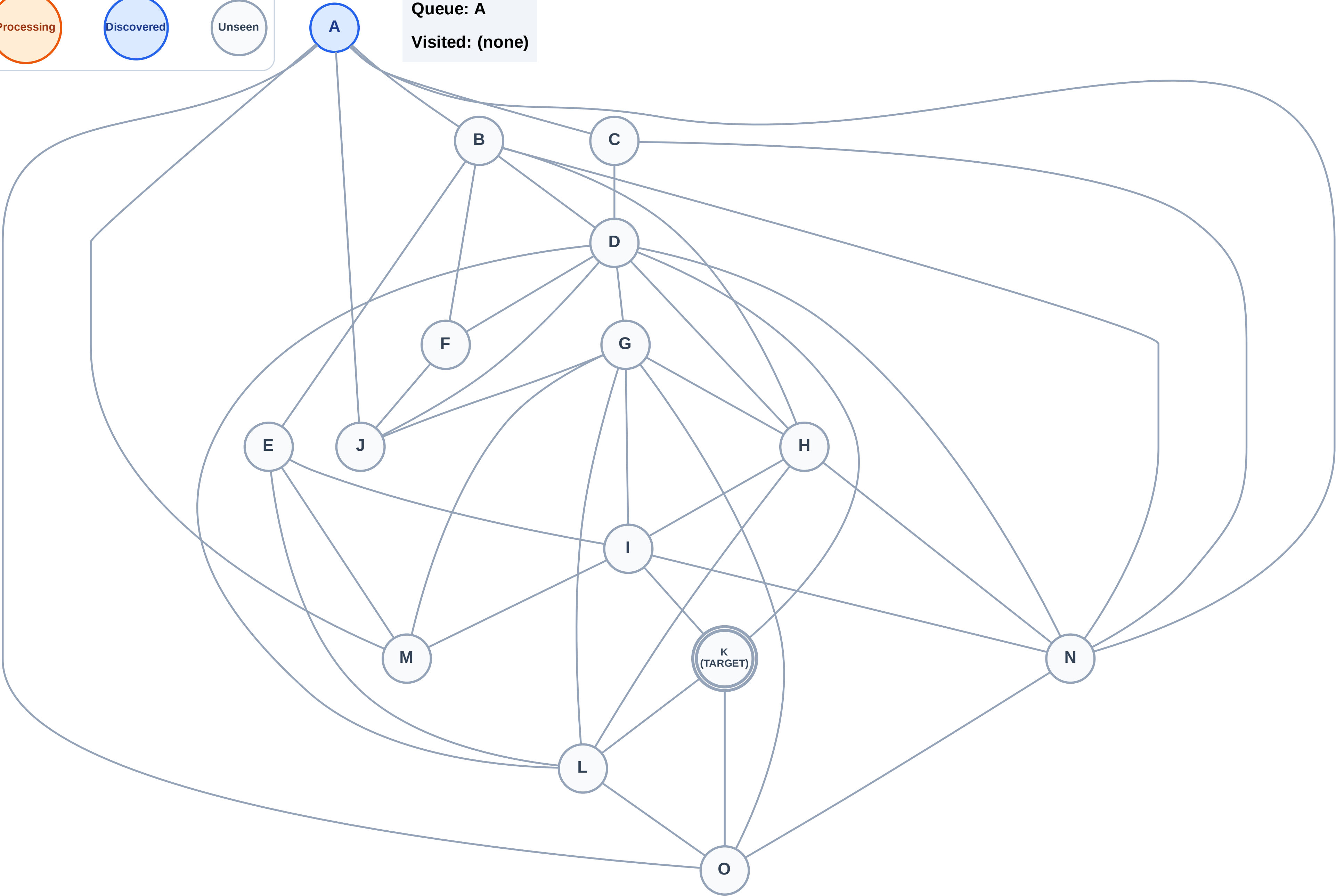
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

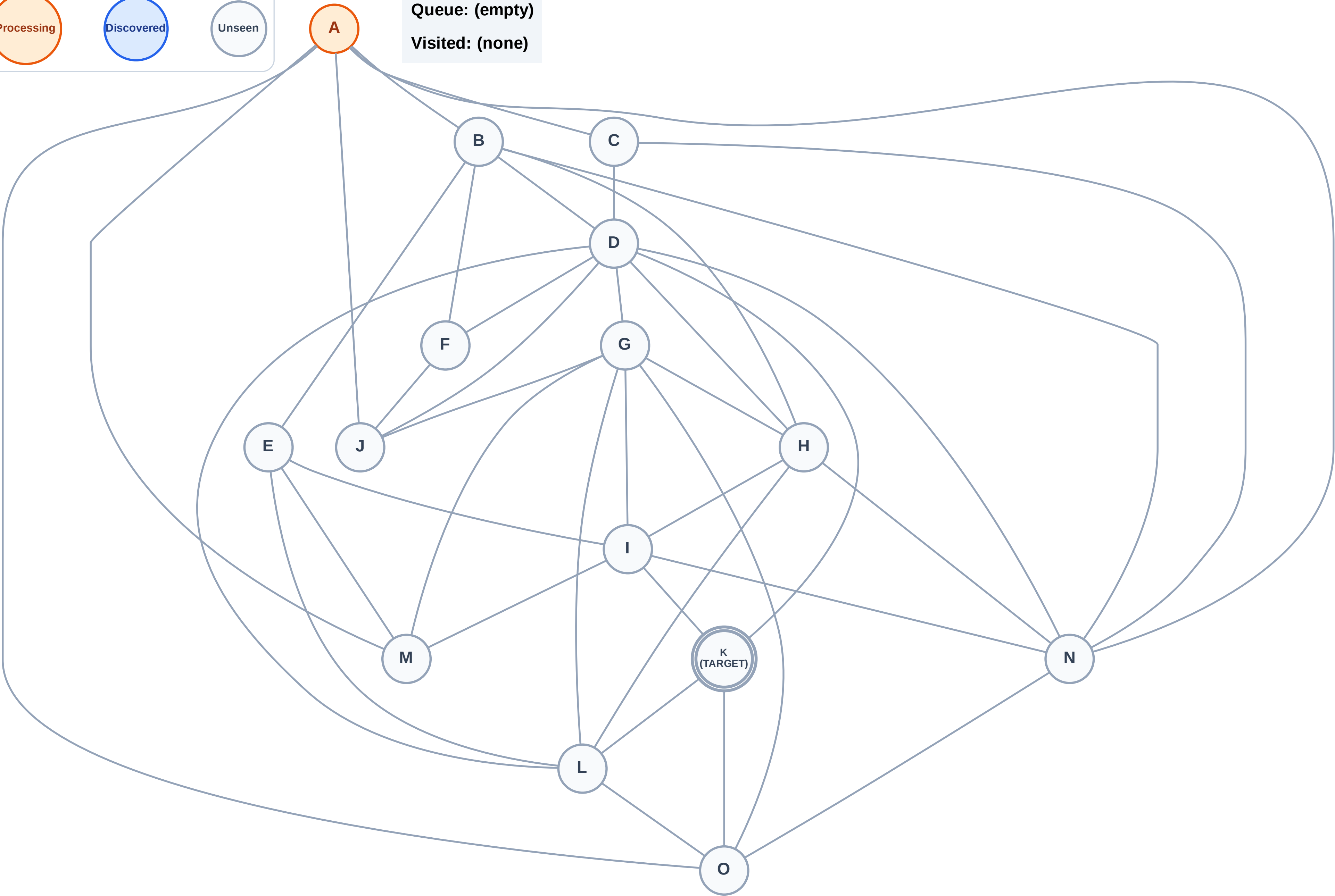
Complete

Processing

Discovered

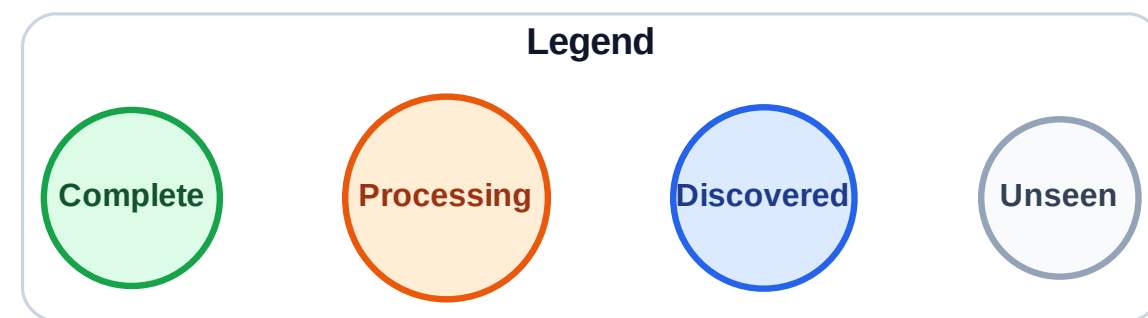
Unseen

Queue: (empty)
Visited: (none)

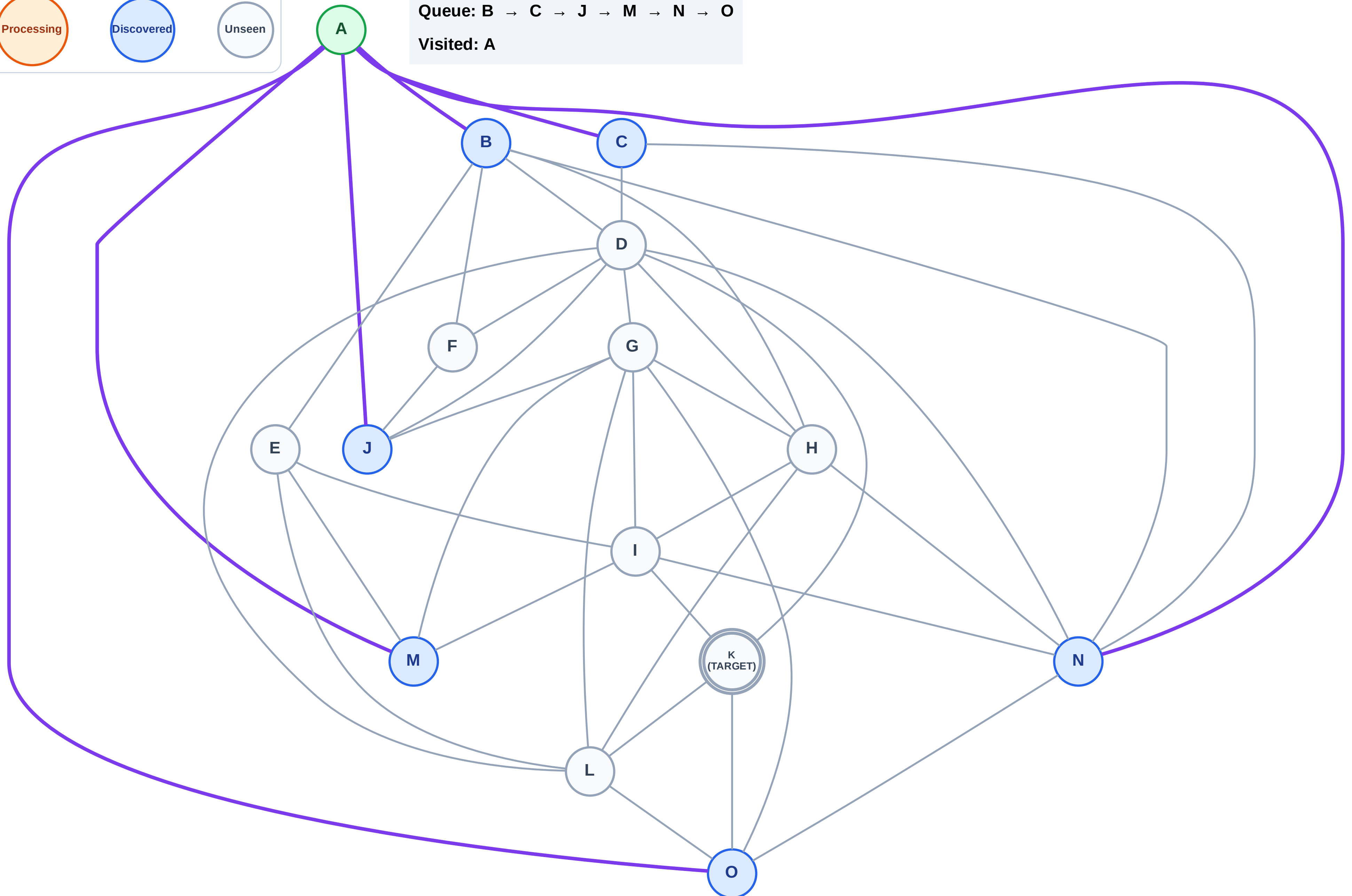


BFS Traversal

Step 3: Discover B, C, J, M, N, O from A



Queue: B → C → J → M → N → O
Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

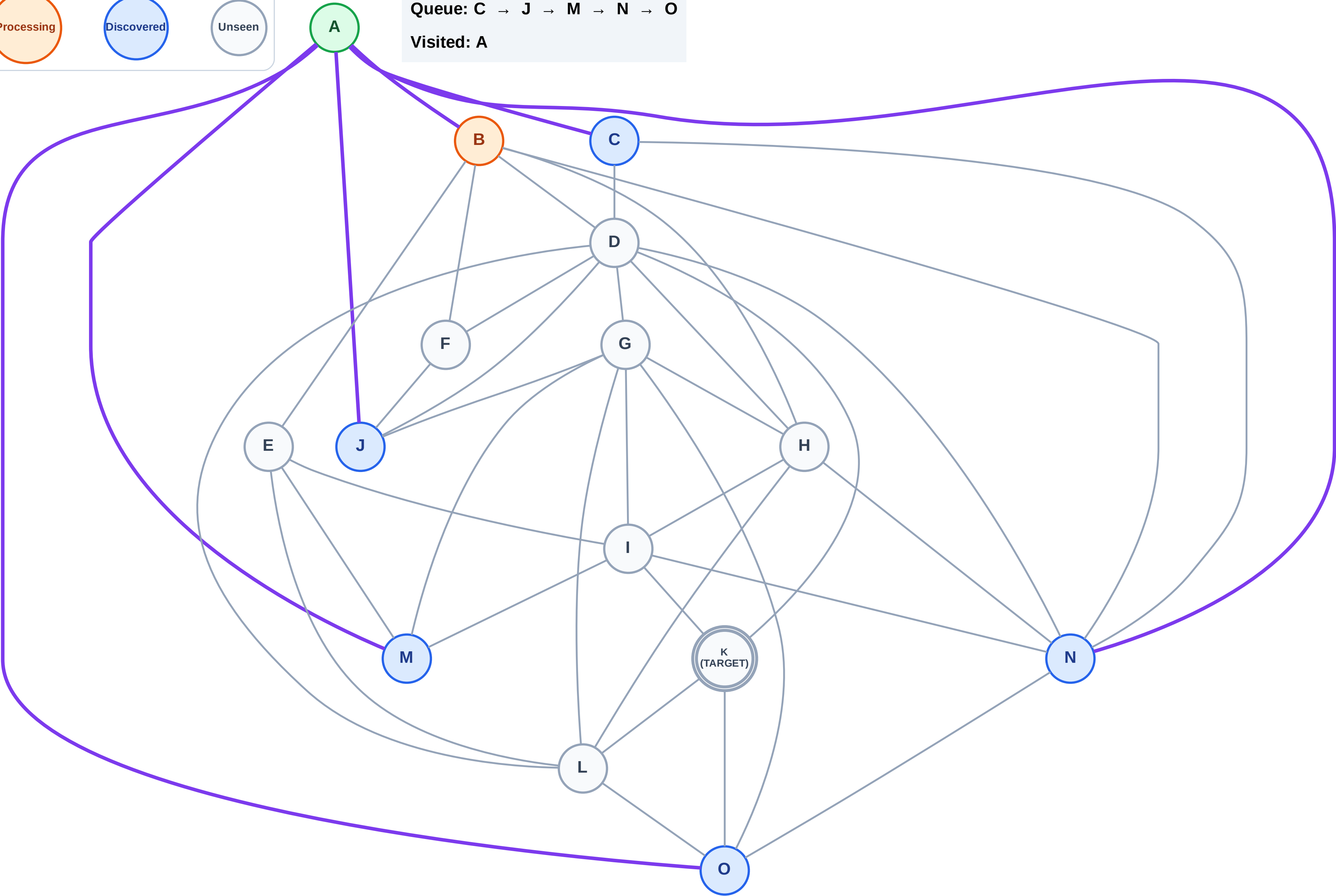
Processing

Discovered

Unseen

Queue: C → J → M → N → O

Visited: A



BFS Traversal

Step 5: Discover D, E, F, H from B

Legend

Complete

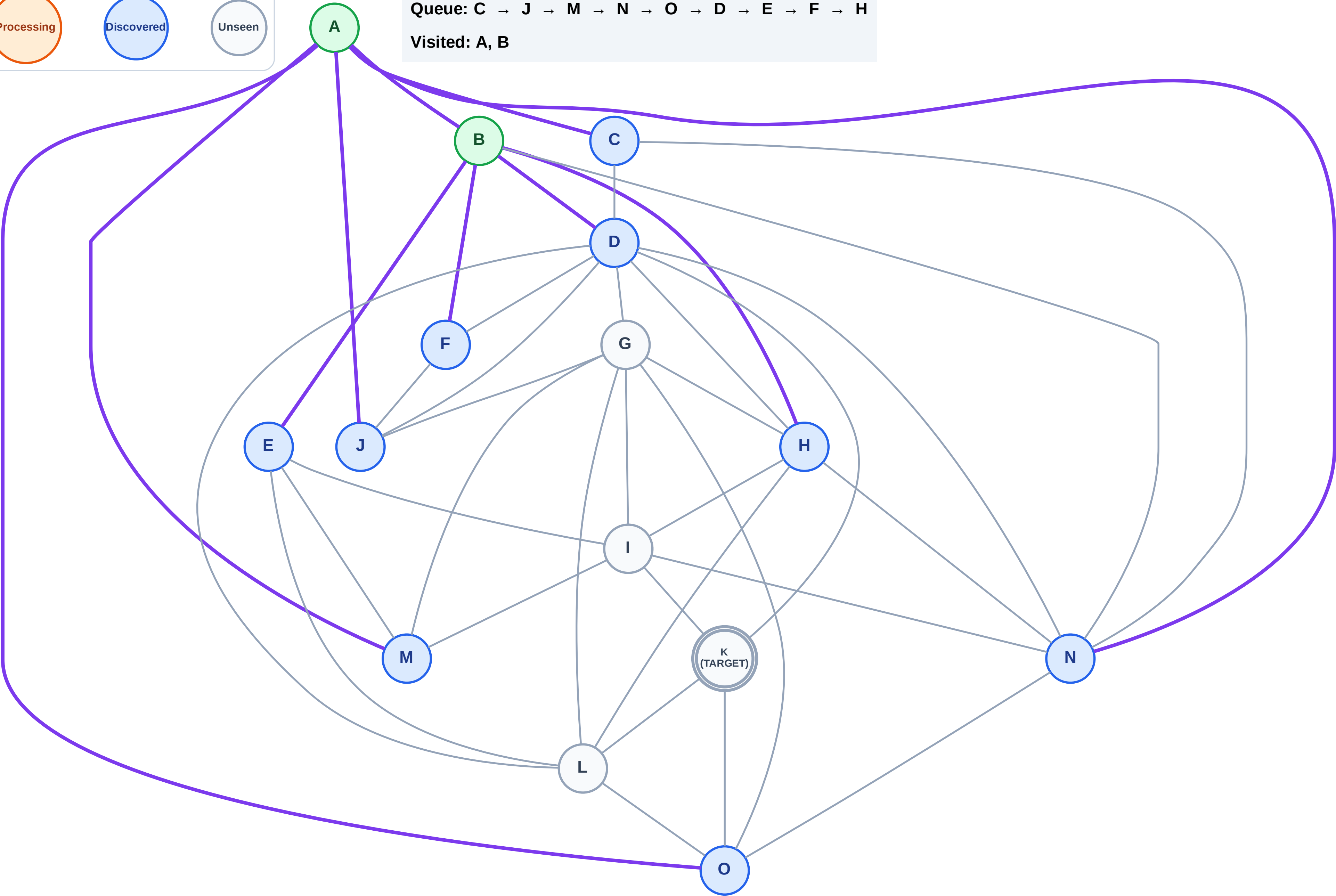
Processing

Discovered

Unseen

Queue: C → J → M → N → O → D → E → F → H

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

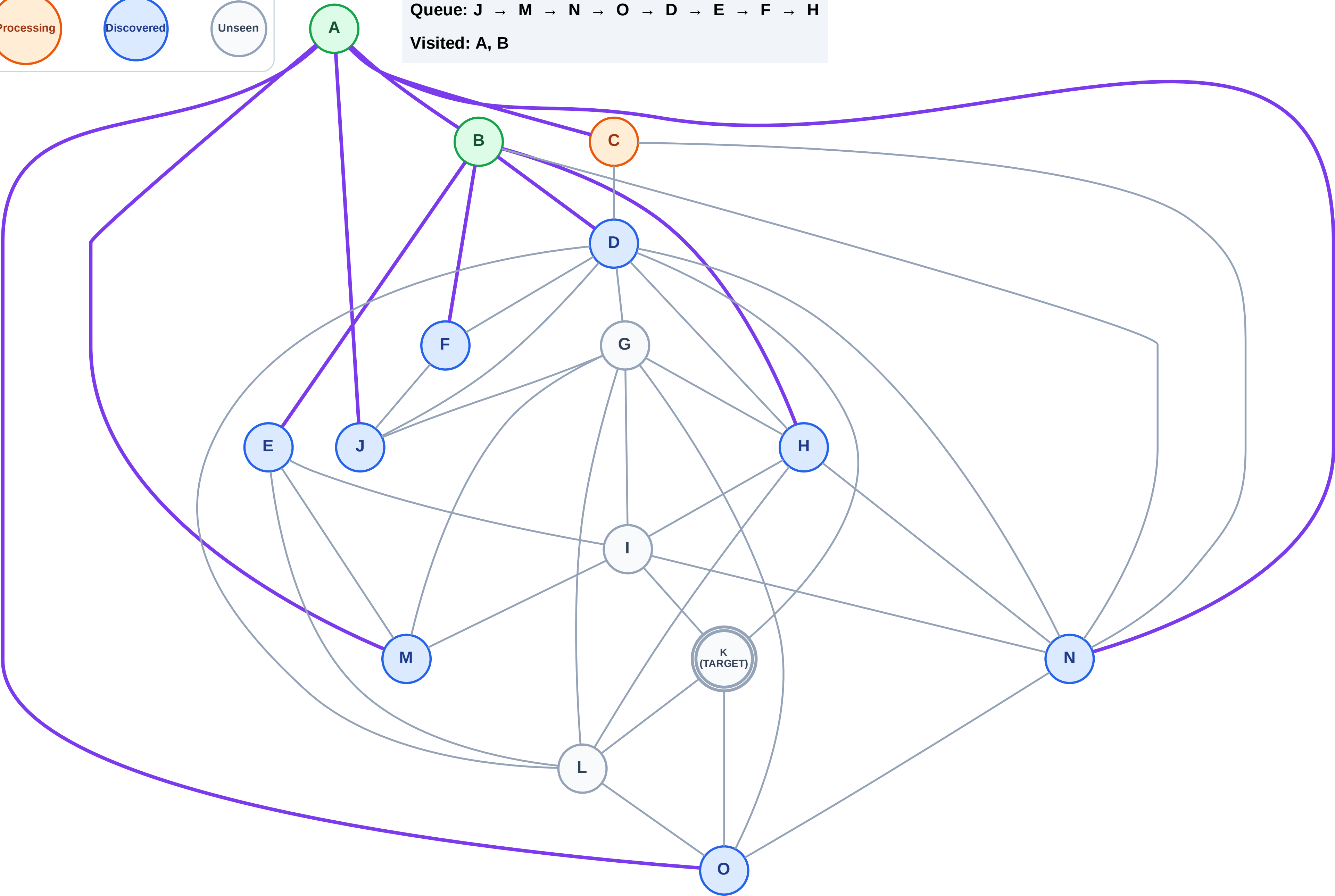
Processing

Discovered

Unseen

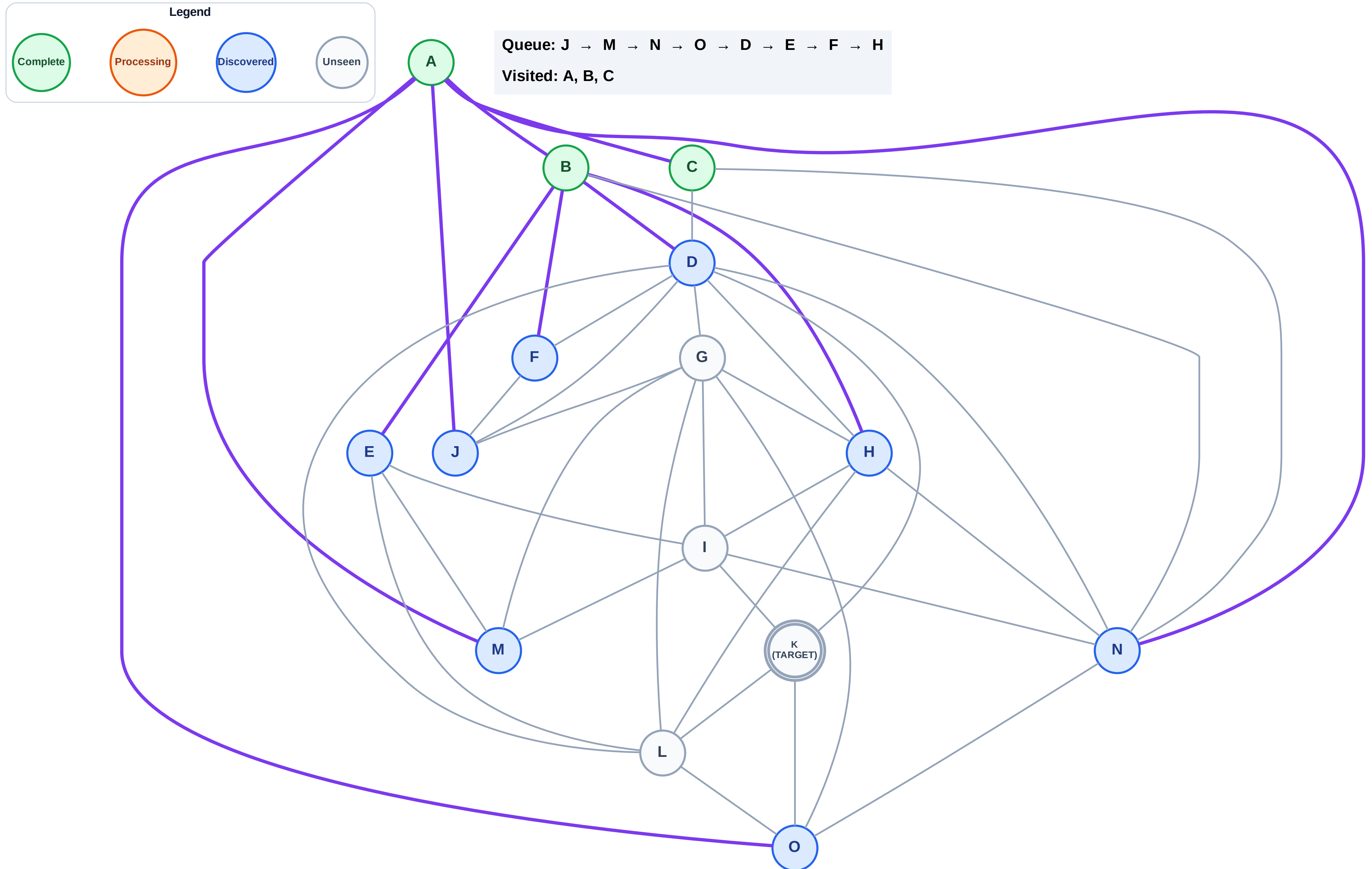
Queue: J → M → N → O → D → E → F → H

Visited: A, B



Step 7: No new neighbors from C

Step 7: No new neighbors from C



BFS Traversal

Step 8: Process node J

Legend

Complete

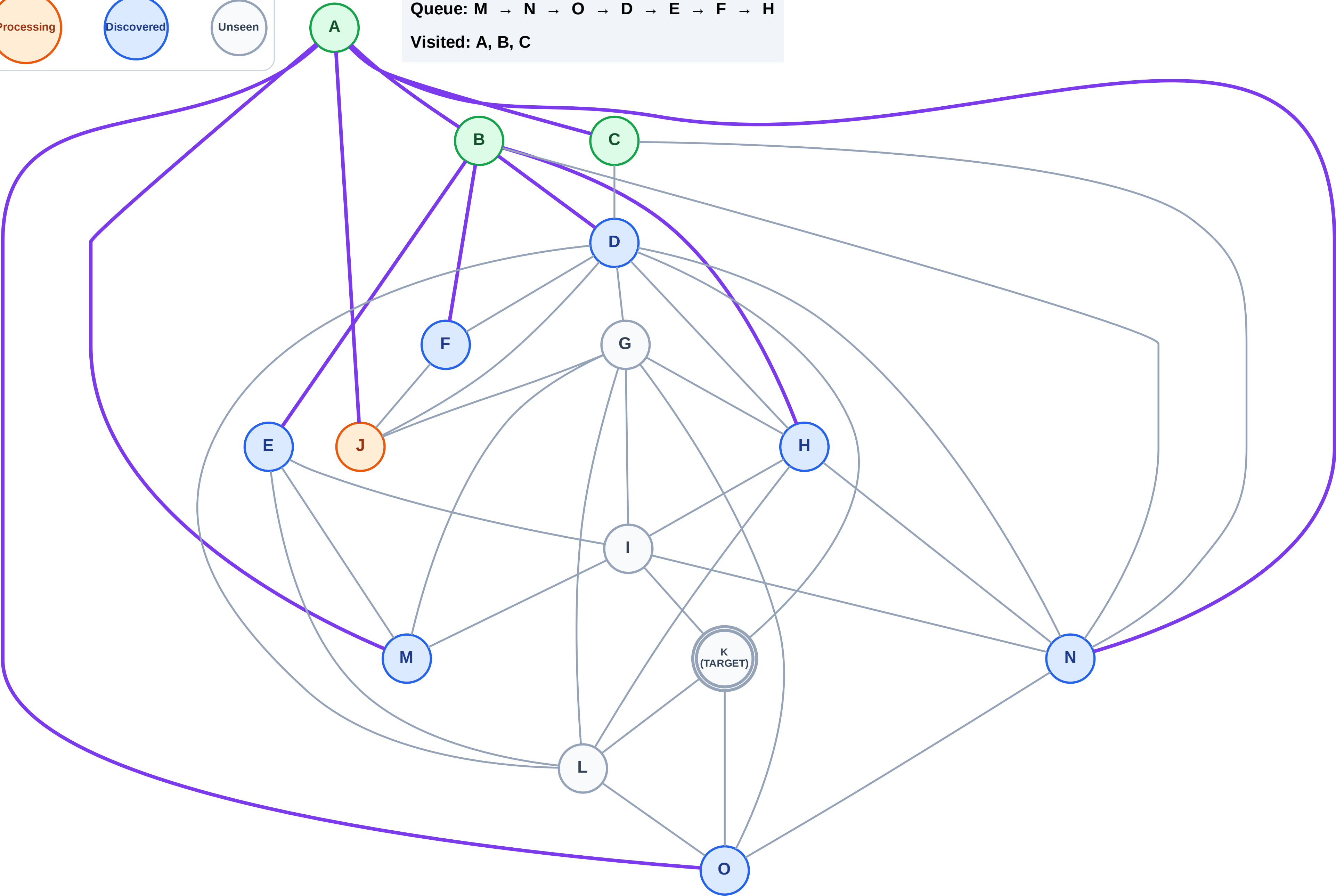
Processing

Discovered

Unseen

Queue: M → N → O → D → E → F → H

Visited: A, B, C



BFS Traversal

Step 9: Discover G from J

Legend

Complete

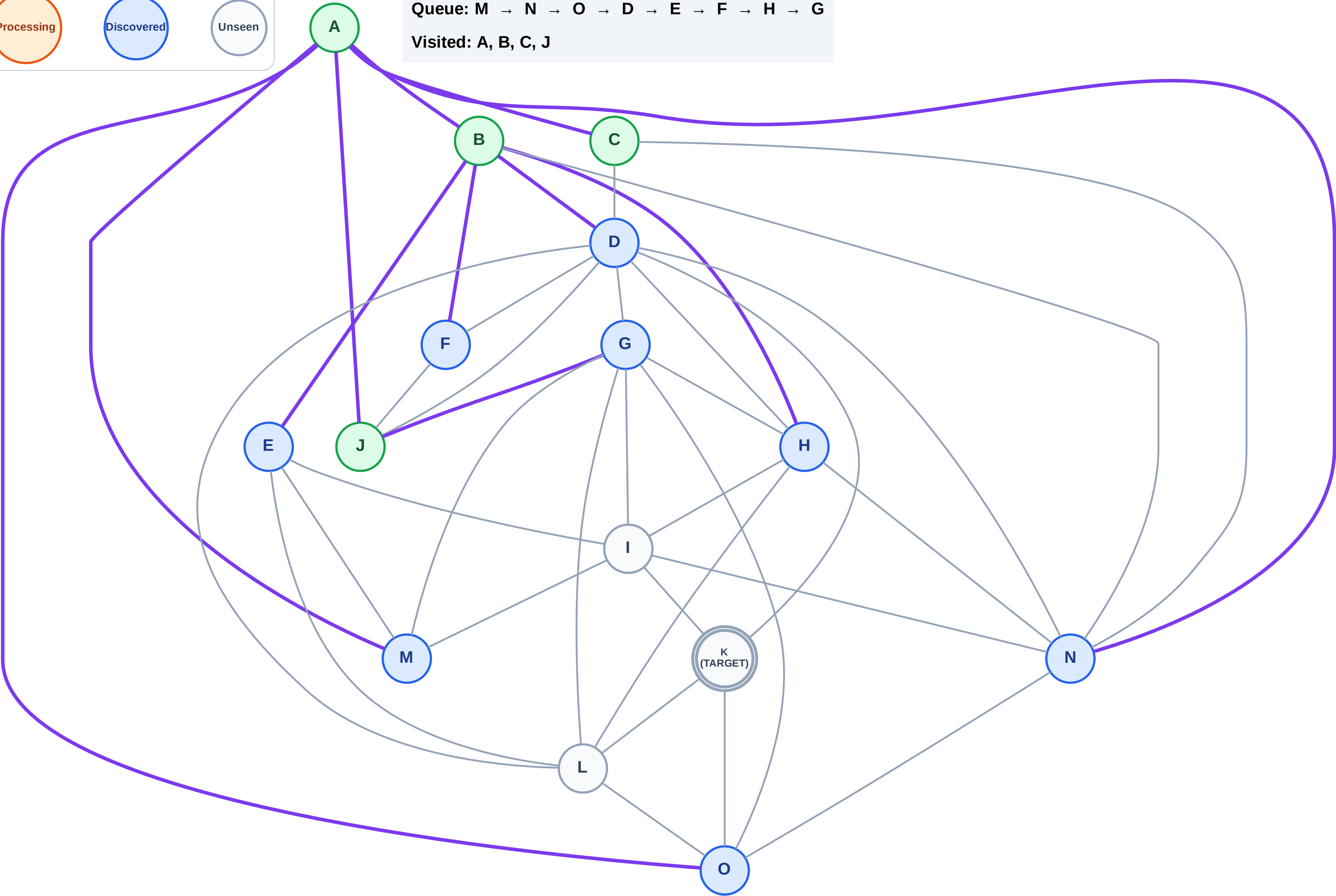
Processing

Discovered

Unseen

Queue: M → N → O → D → E → F → H → G

Visited: A, B, C, J



BFS Traversal

Step 10: Process node M

Legend

Complete

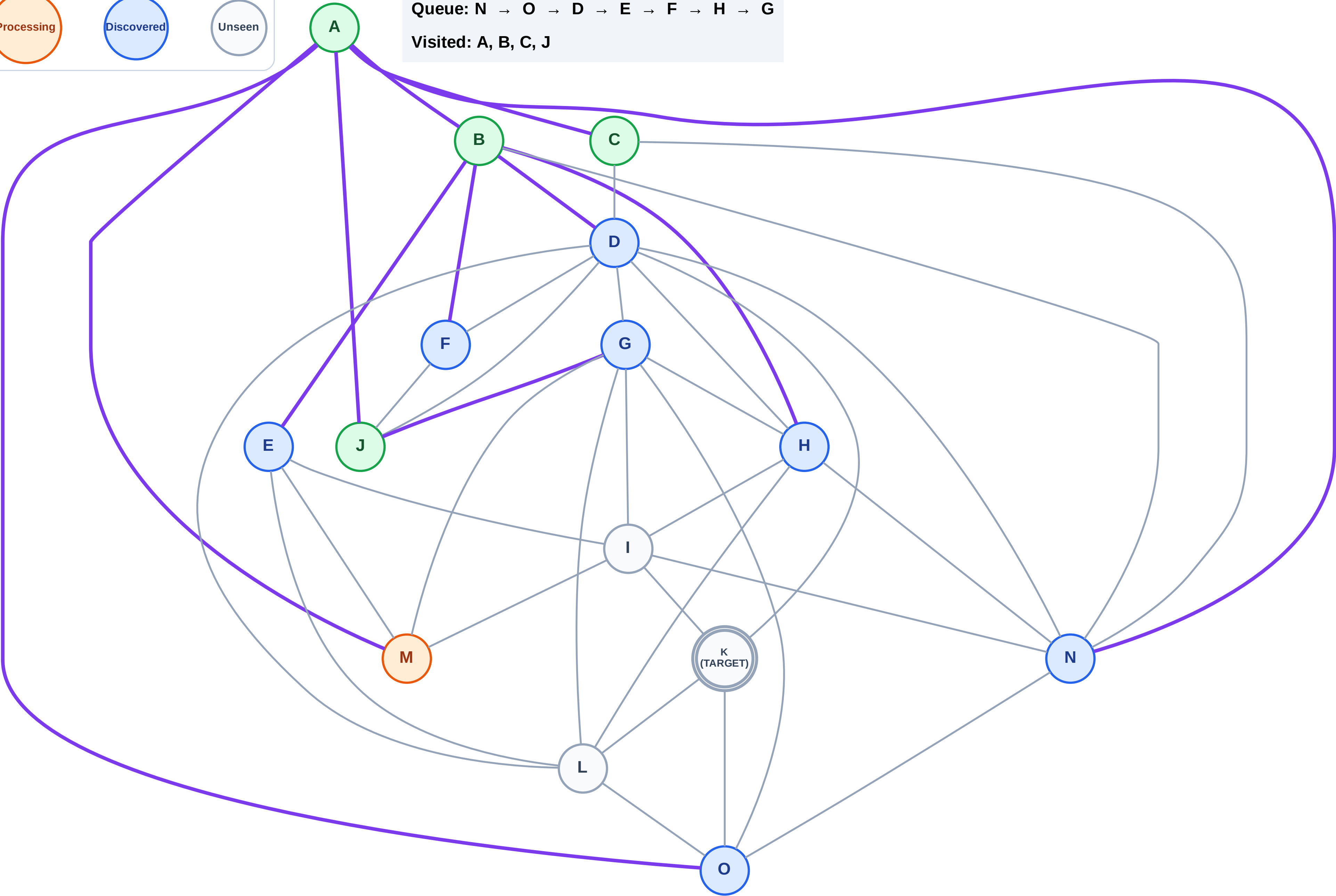
Processing

Discovered

Unseen

Queue: N → O → D → E → F → H → G

Visited: A, B, C, J



Step 11: Discover I from M

Complete



Processing

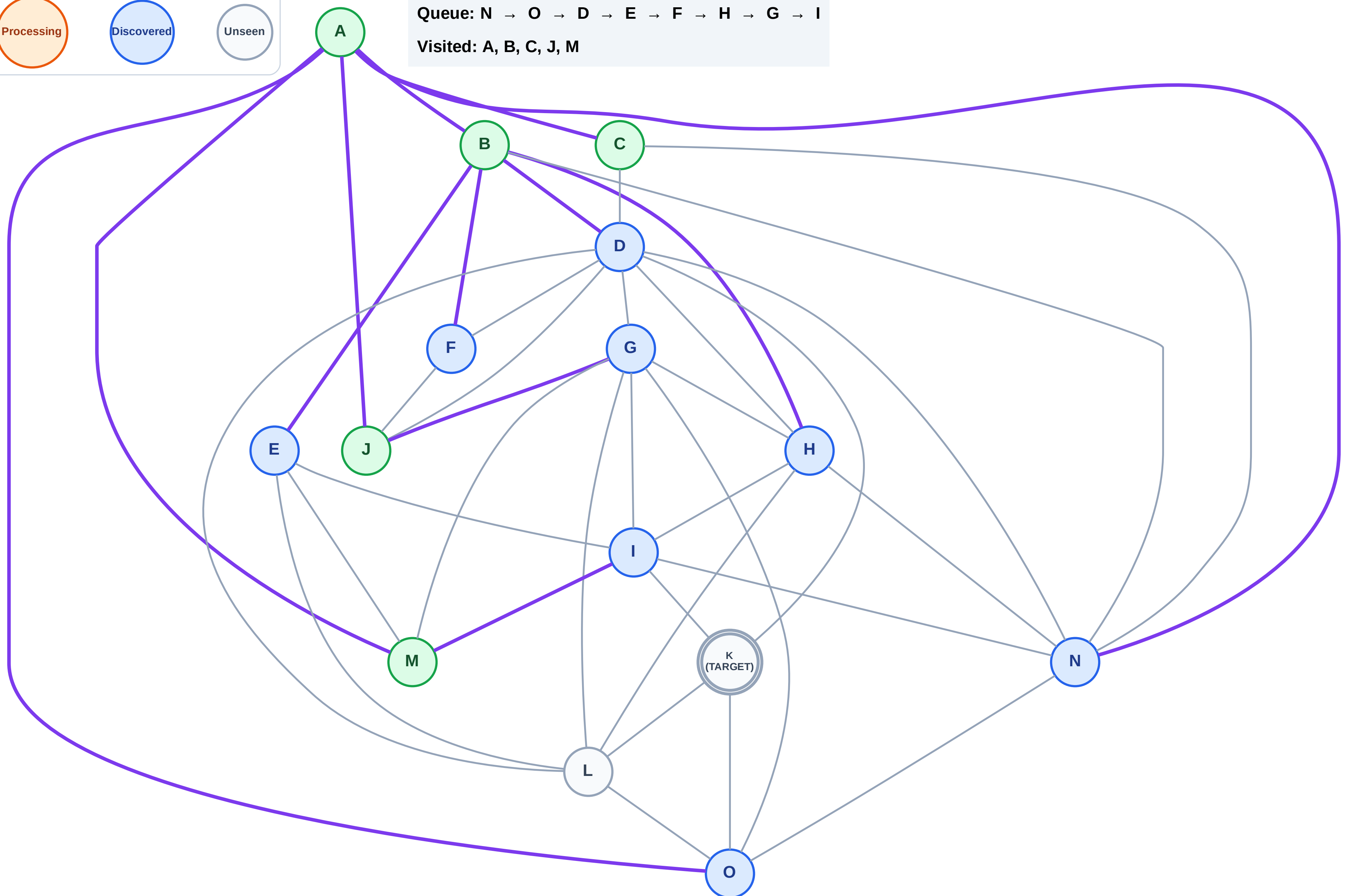


Discovered

Unseen

Unseen

Visited: A, B, C, J, M



BFS Traversal

Step 12: Process node N

Legend

Complete

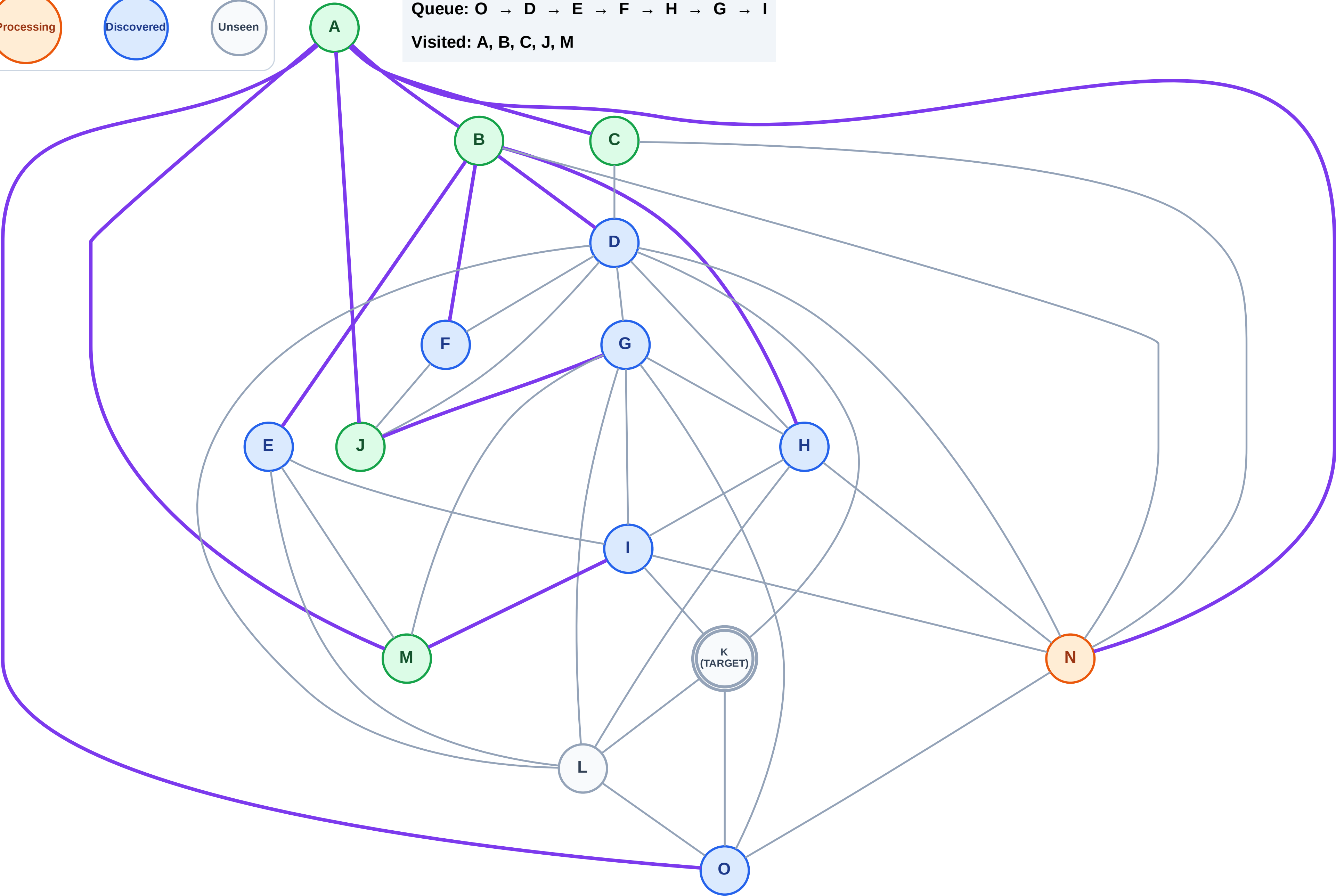
Processing

Discovered

Unseen

Queue: O → D → E → F → H → G → I

Visited: A, B, C, J, M



BFS Traversal

Step 13: No new neighbors from N

Legend

Complete

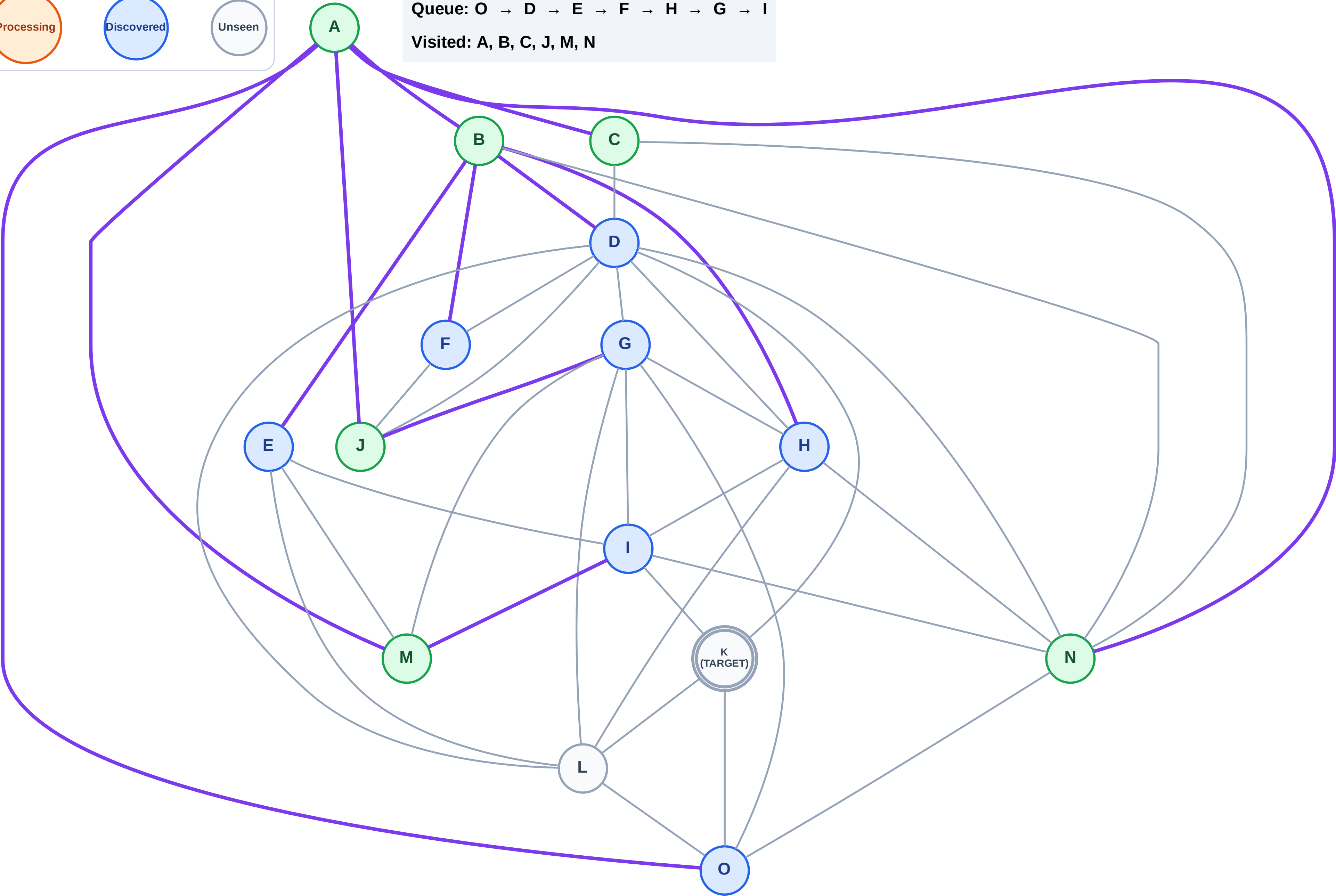
Processing

Discovered

Unseen

Queue: O → D → E → F → H → G → I

Visited: A, B, C, J, M, N



BFS Traversal

Step 14: Process node O

Legend

Complete

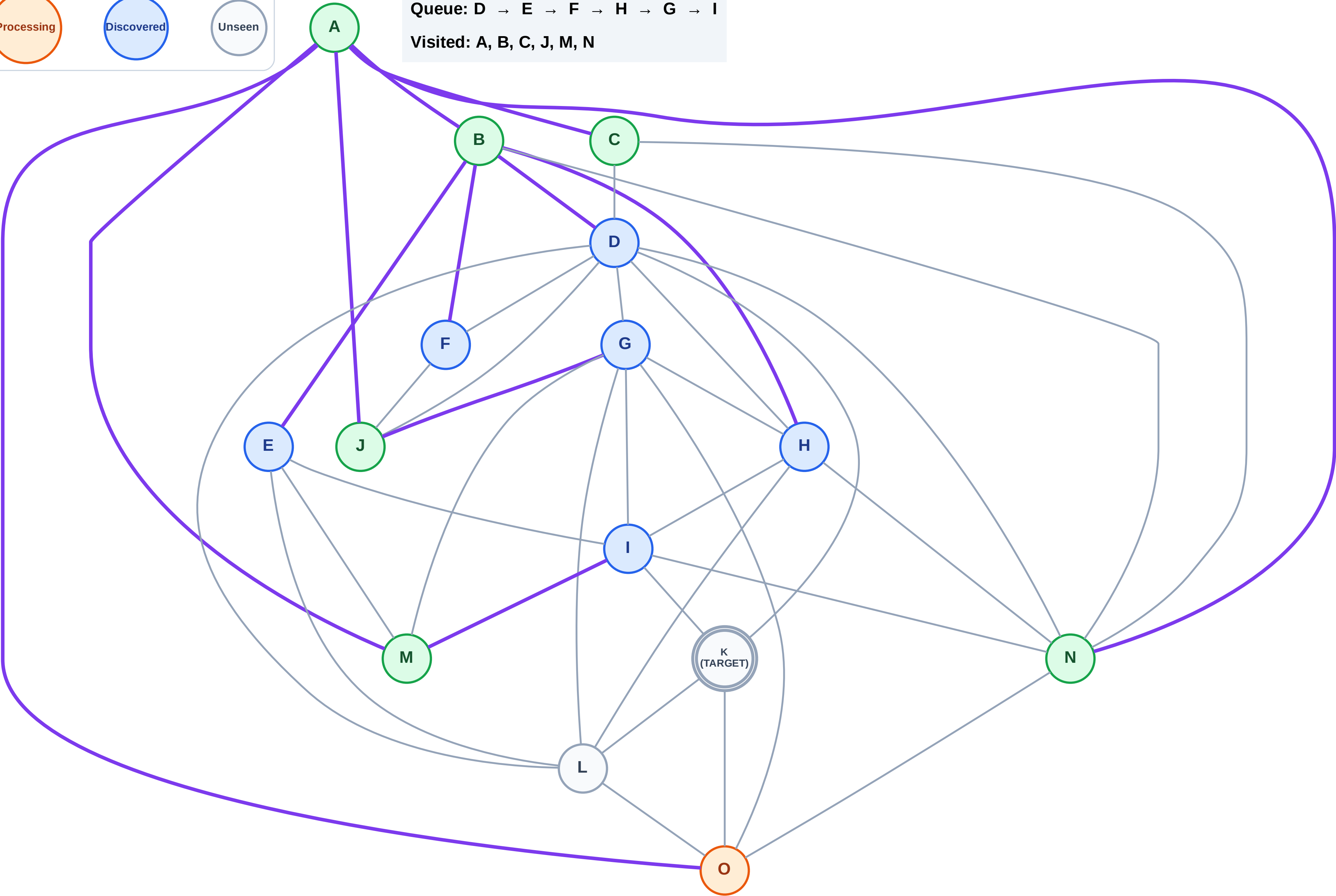
Processing

Discovered

Unseen

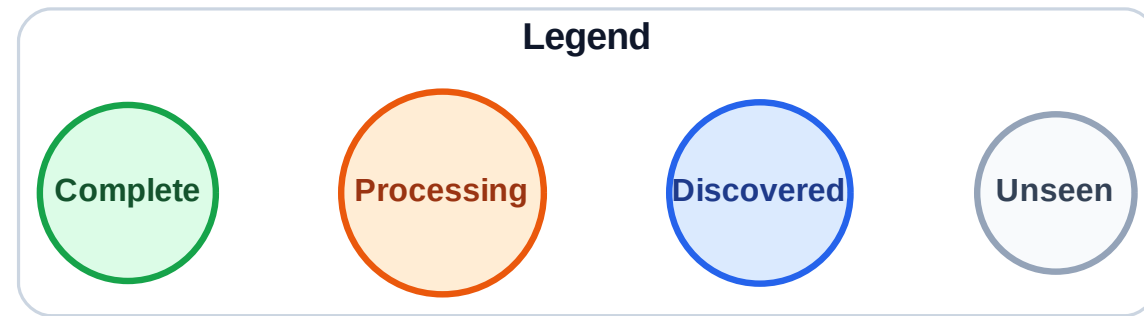
Queue: D → E → F → H → G → I

Visited: A, B, C, J, M, N

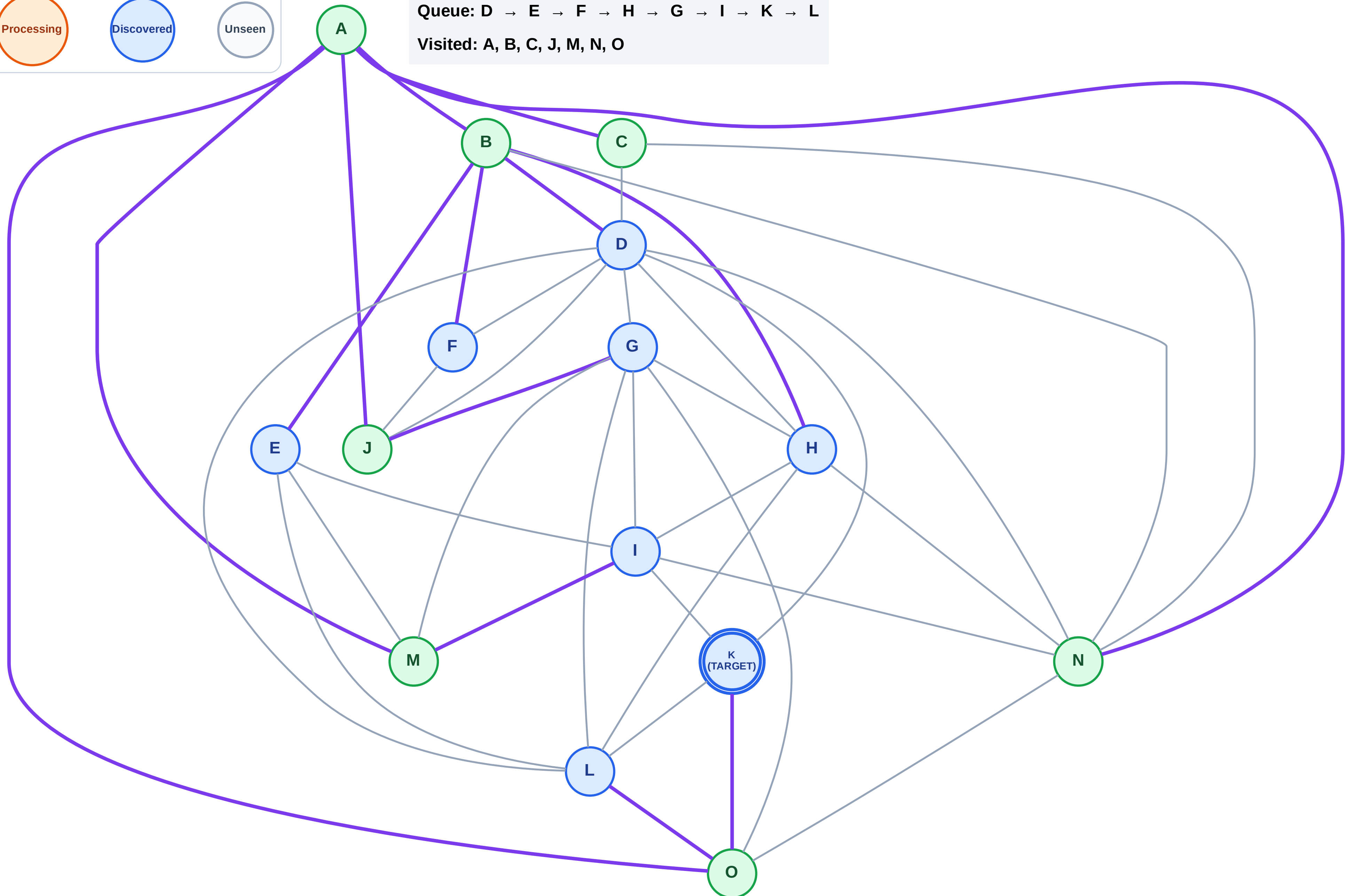


BFS Traversal

Step 15: Discover K, L from O



Queue: D → E → F → H → G → I → K → L
Visited: A, B, C, J, M, N, O



BFS Traversal

Step 16: Process node D

Legend

Complete

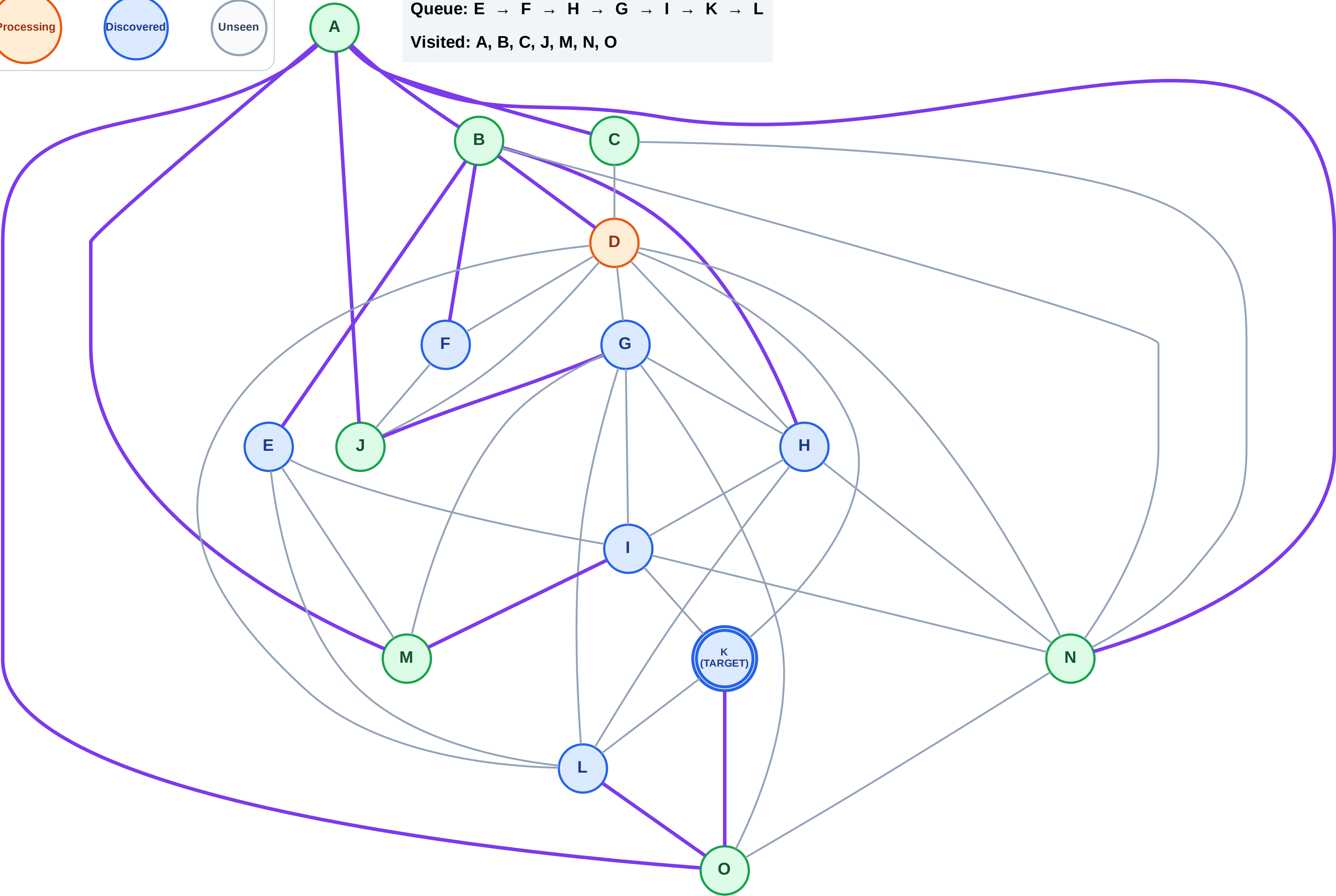
Processing

Discovered

Unseen

Queue: E → F → H → G → I → K → L

Visited: A, B, C, J, M, N, O



BFS Traversal

Step 17: No new neighbors from D

Legend

Complete

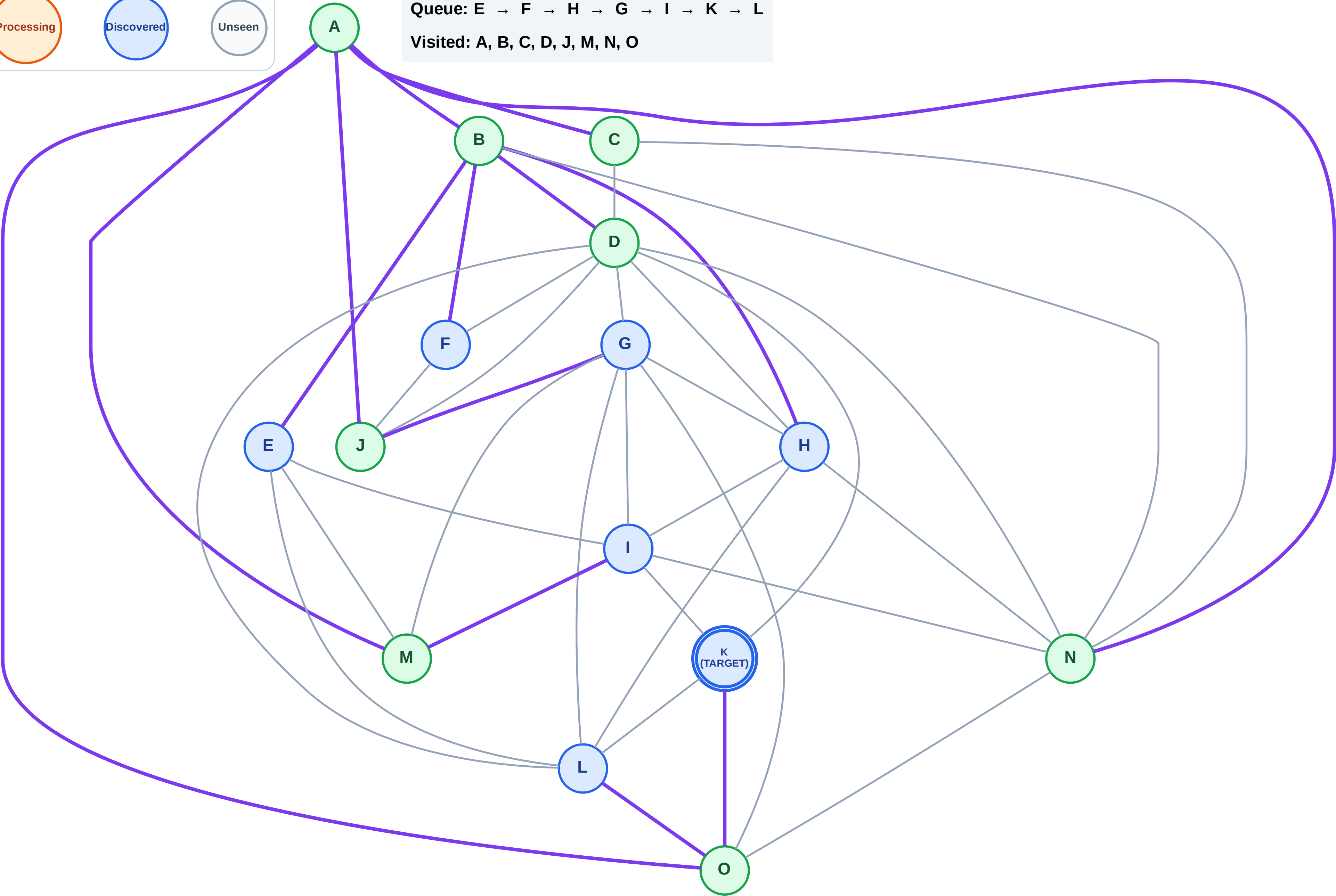
Processing

Discovered

Unseen

Queue: E → F → H → G → I → K → L

Visited: A, B, C, D, J, M, N, O



BFS Traversal

Step 18: Process node E

Legend

Complete

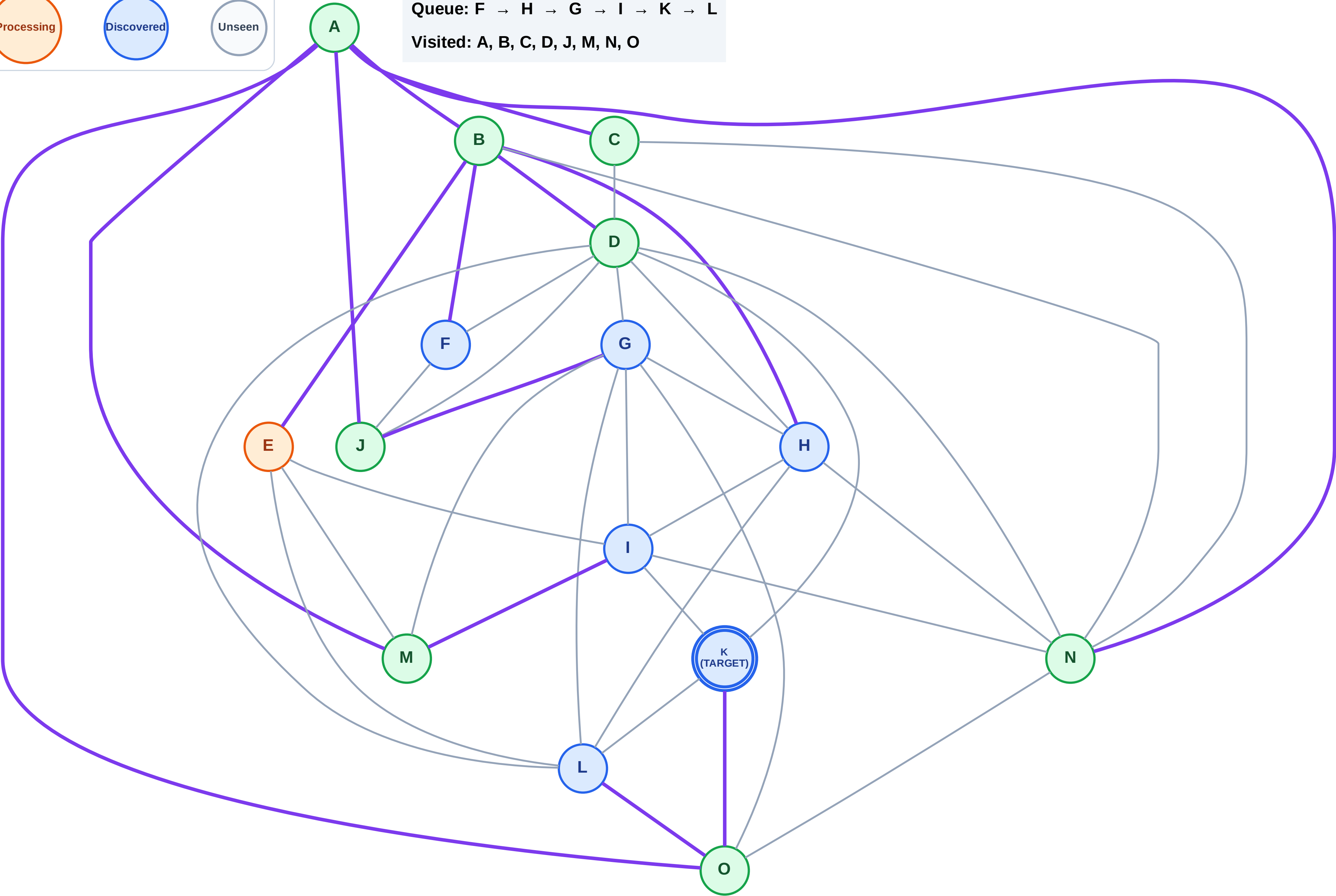
Processing

Discovered

Unseen

Queue: F → H → G → I → K → L

Visited: A, B, C, D, J, M, N, O



BFS Traversal

Step 19: No new neighbors from E

Legend

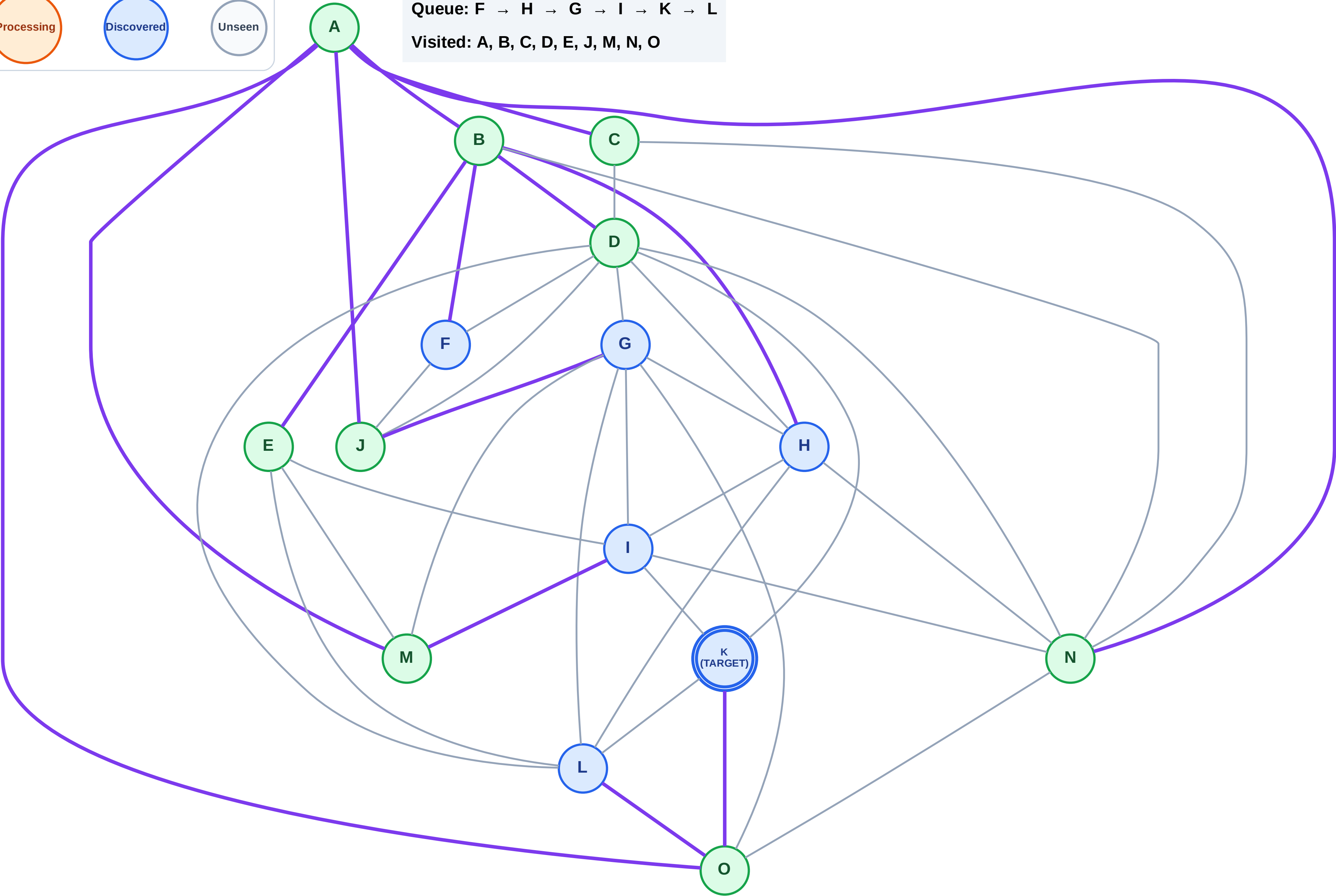
Complete

Processing

Discovered

Unseen

Queue: F → H → G → I → K → L
Visited: A, B, C, D, E, J, M, N, O



BFS Traversal

Step 20: Process node F

Legend

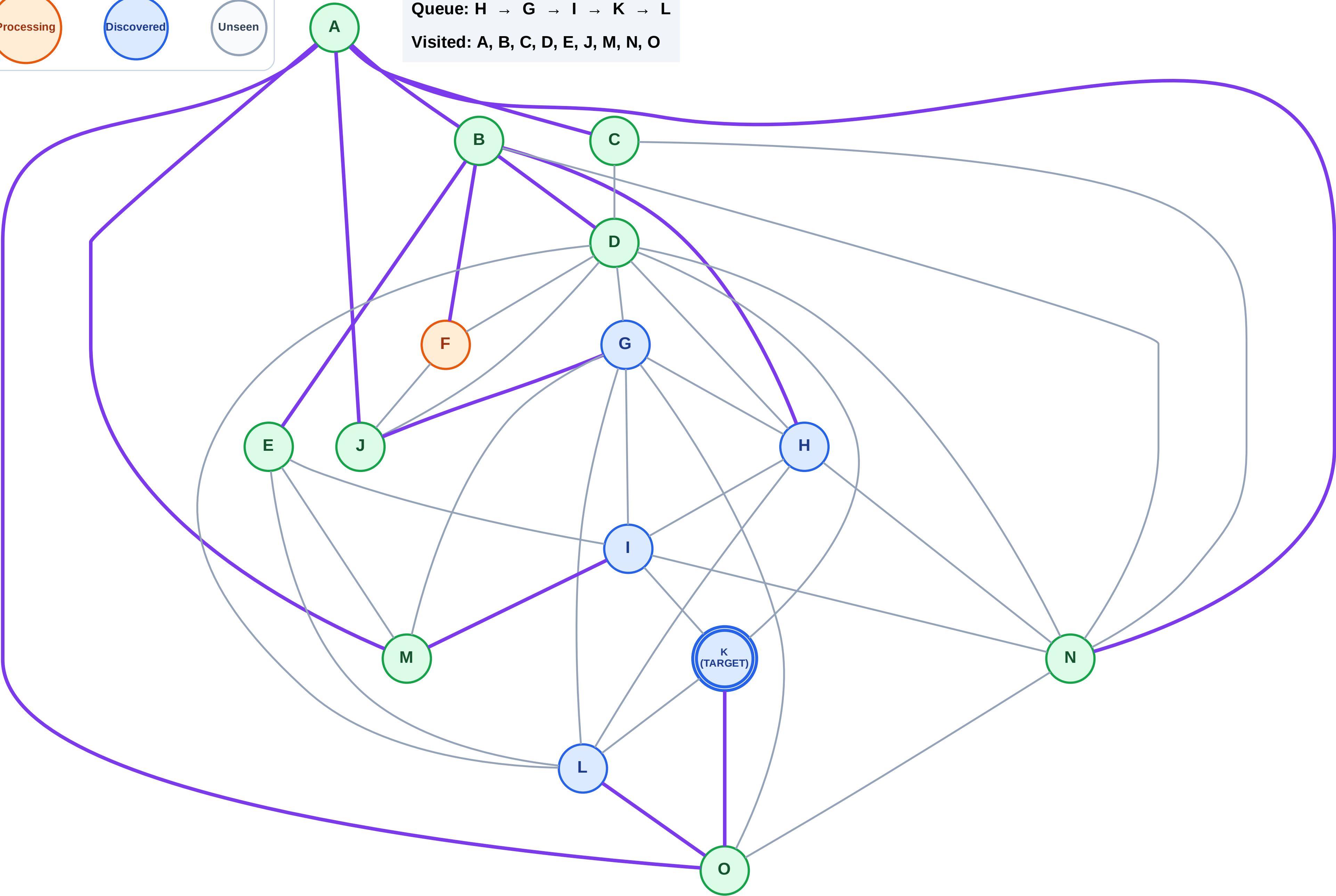
Complete

Processing

Discovered

Unseen

Queue: H → G → I → K → L
Visited: A, B, C, D, E, J, M, N, O



BFS Traversal

Step 21: No new neighbors from F

Legend

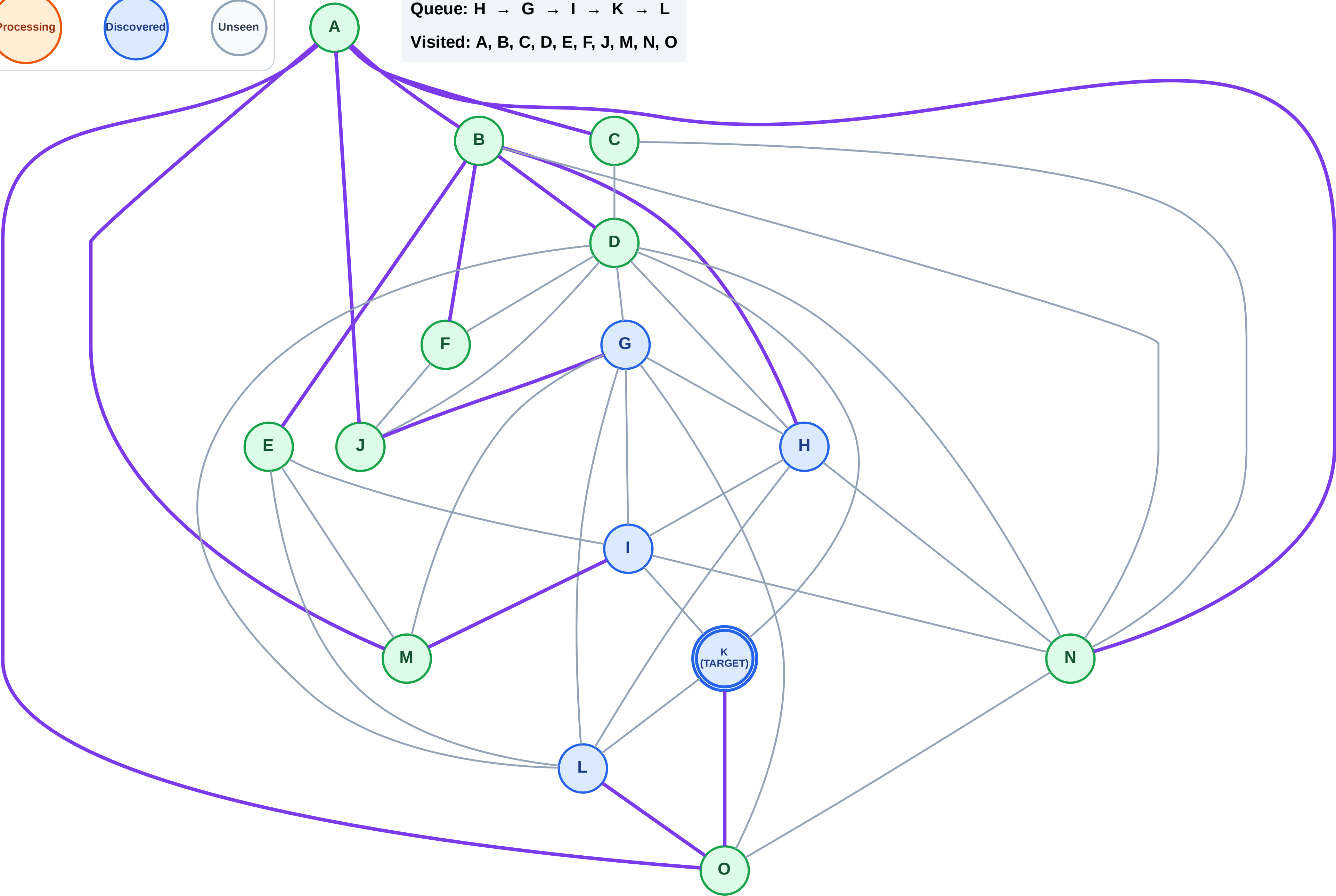
Complete

Processing

Discovered

Unseen

Queue: H → G → I → K → L
Visited: A, B, C, D, E, F, J, M, N, O



BFS Traversal

Step 22: Process node H

Legend

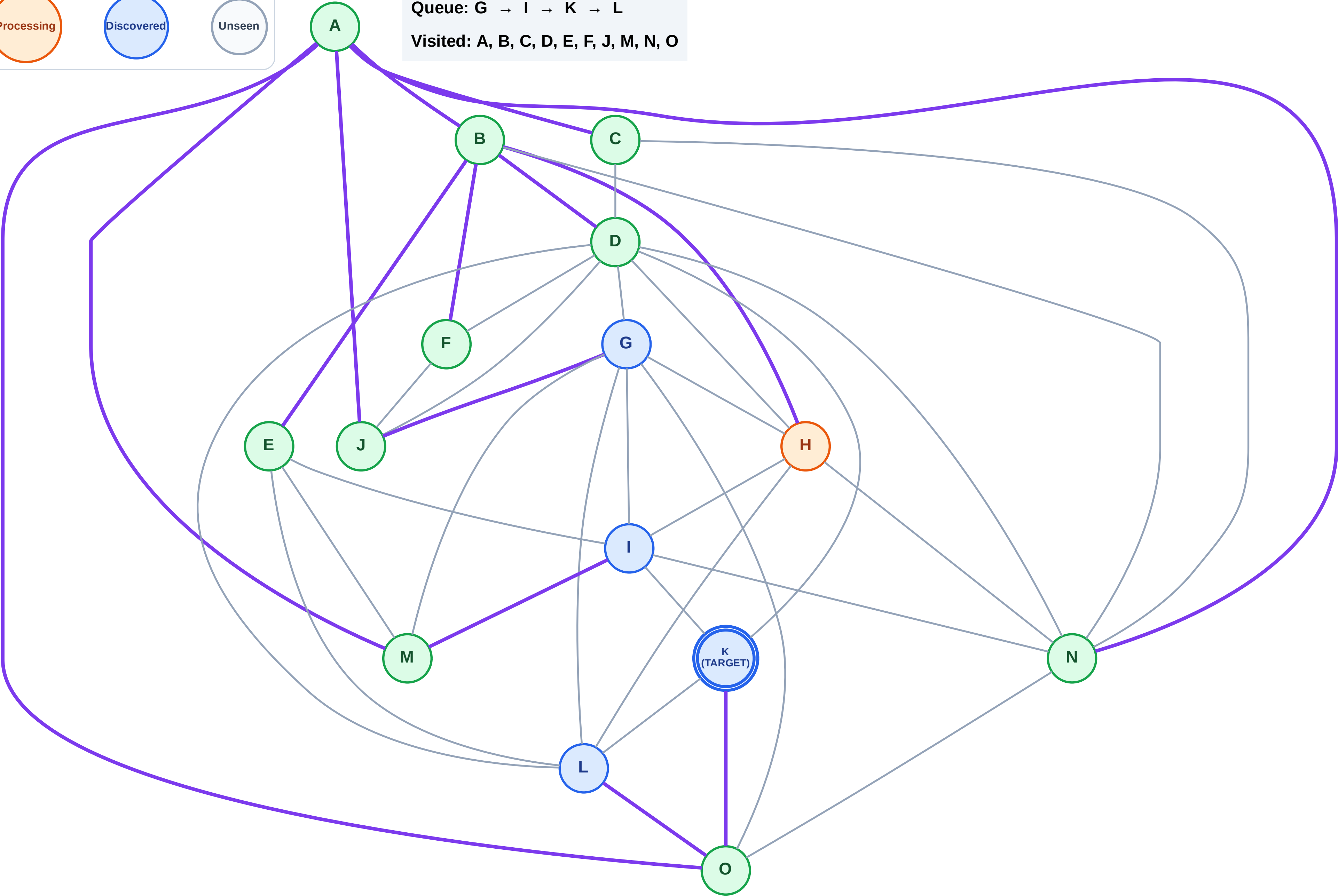
Complete

Processing

Discovered

Unseen

Queue: G → I → K → L
Visited: A, B, C, D, E, F, J, M, N, O



BFS Traversal

Step 23: No new neighbors from H

Legend

Complete

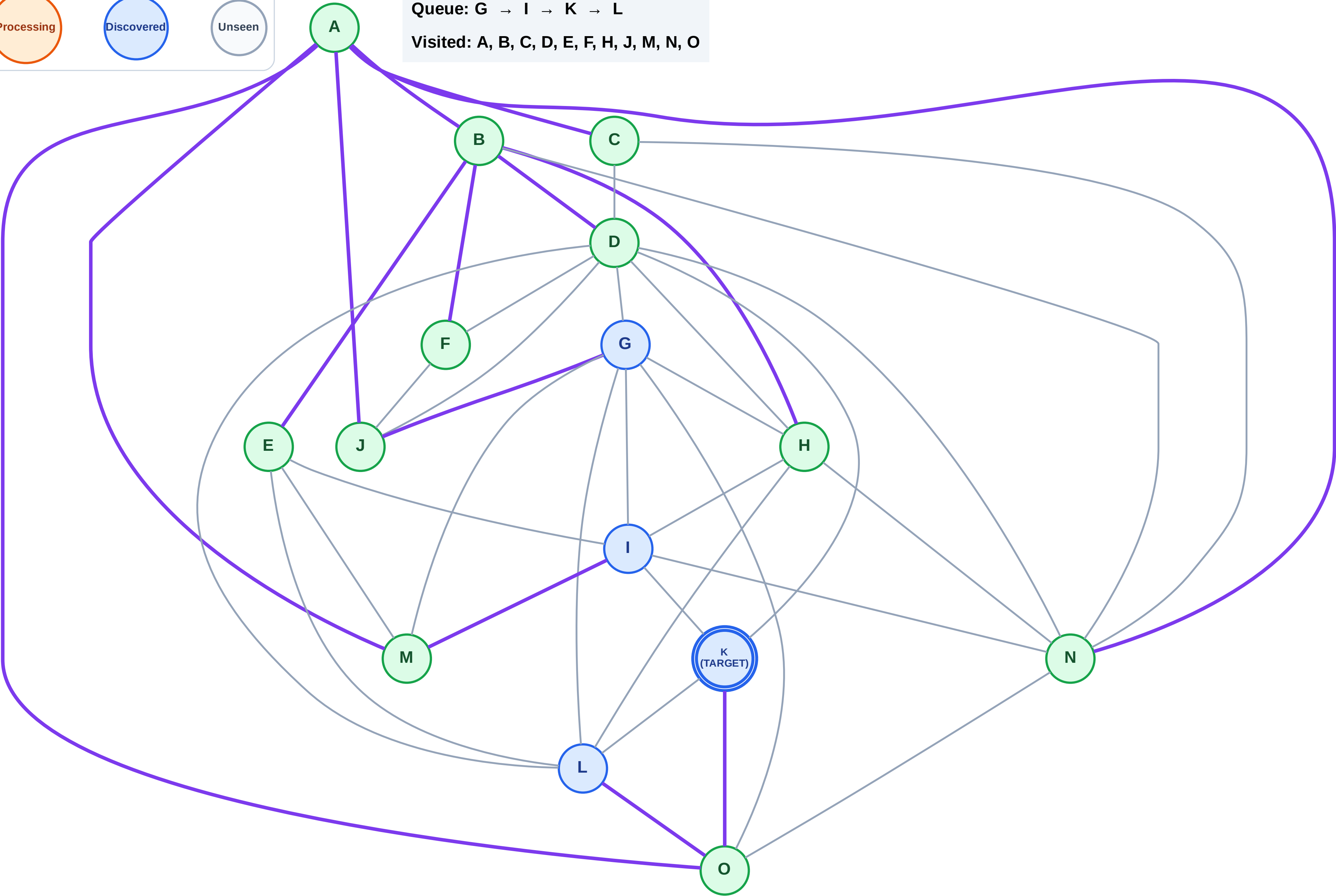
Processing

Discovered

Unseen

Queue: G → I → K → L

Visited: A, B, C, D, E, F, H, J, M, N, O



BFS Traversal

Step 24: Process node G

Legend

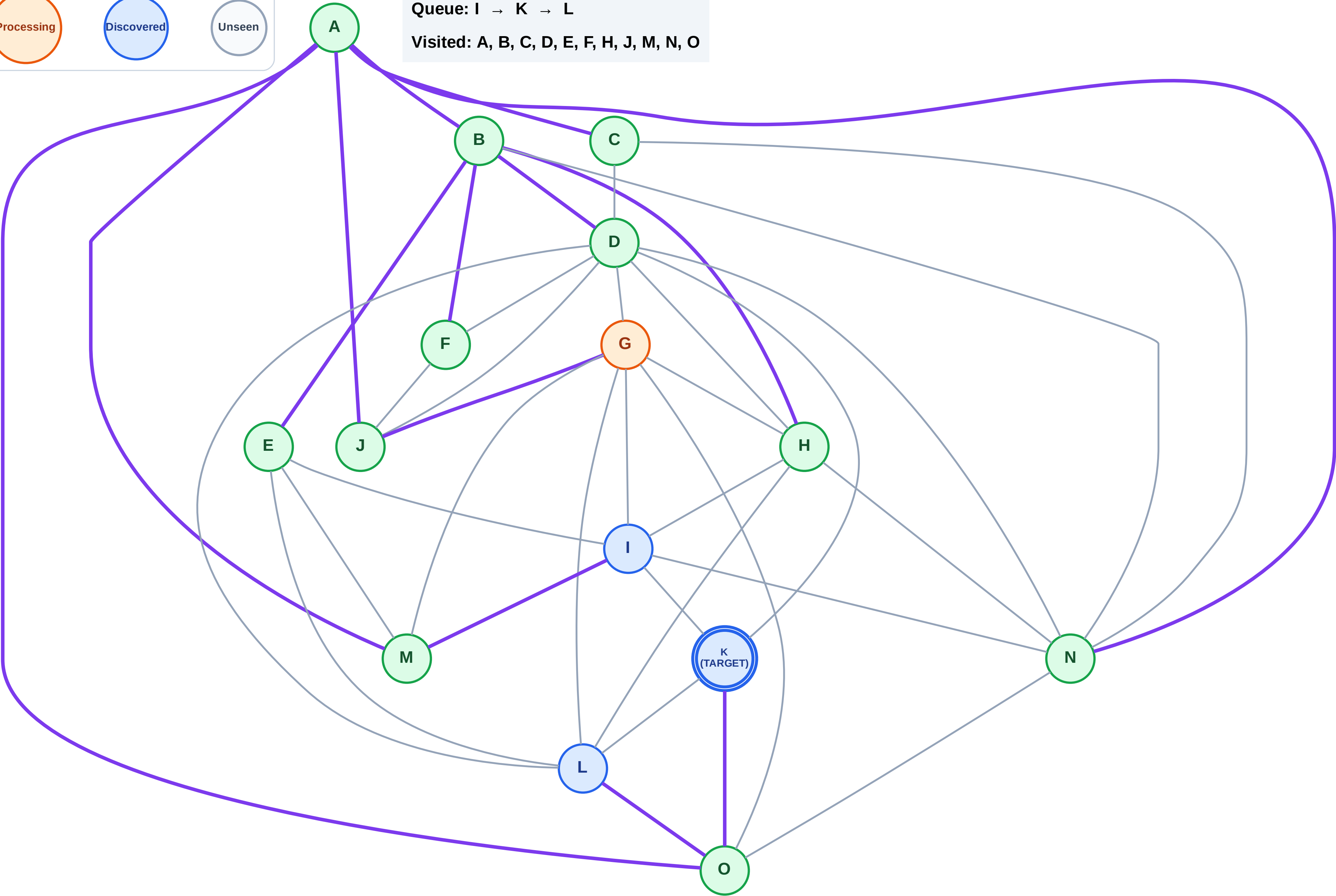
Complete

Processing

Discovered

Unseen

Queue: I → K → L
Visited: A, B, C, D, E, F, H, J, M, N, O



BFS Traversal

Step 25: No new neighbors from G

Legend

Complete

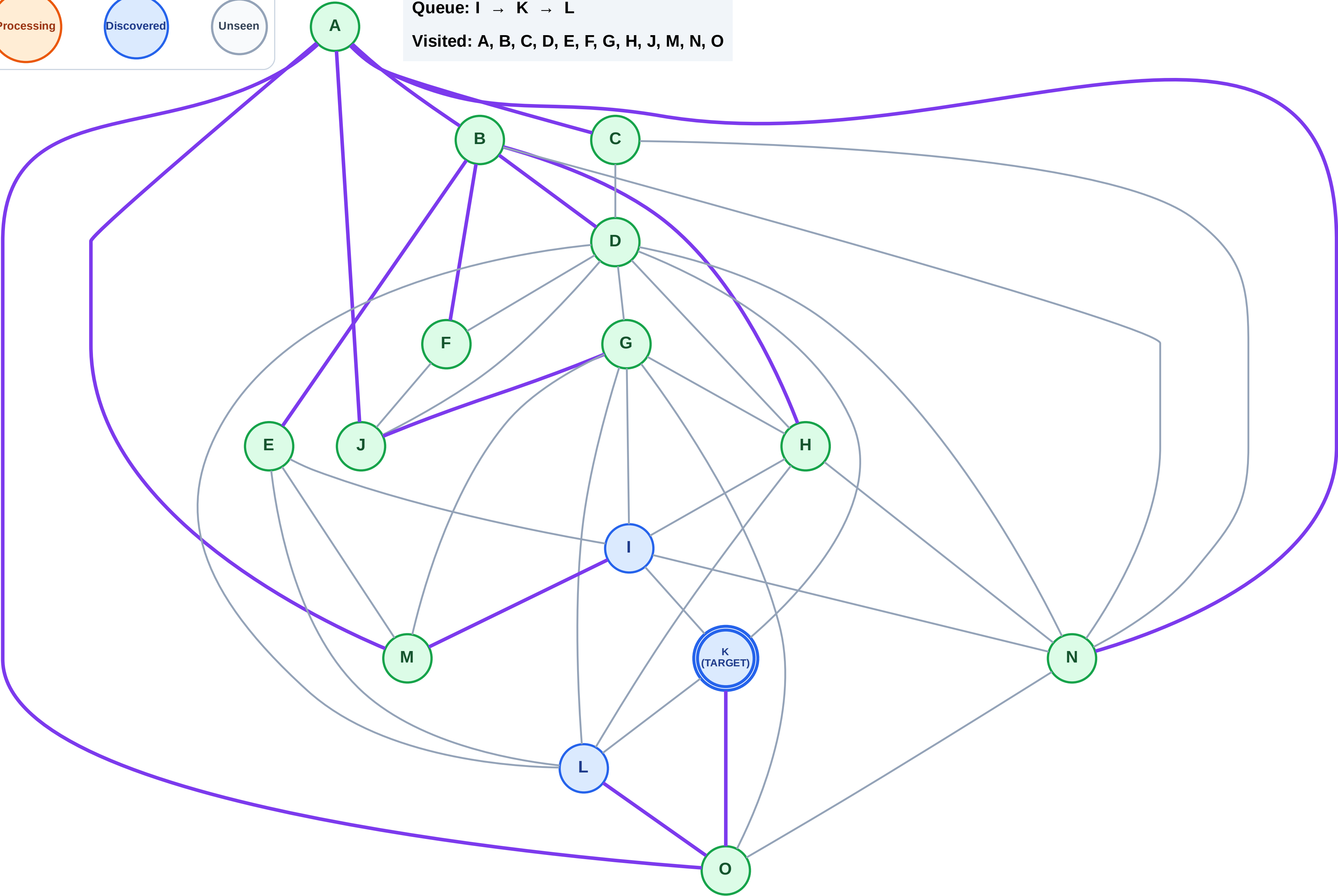
Processing

Discovered

Unseen

Queue: I → K → L

Visited: A, B, C, D, E, F, G, H, J, M, N, O



BFS Traversal

Step 26: Process node I

Legend

Complete

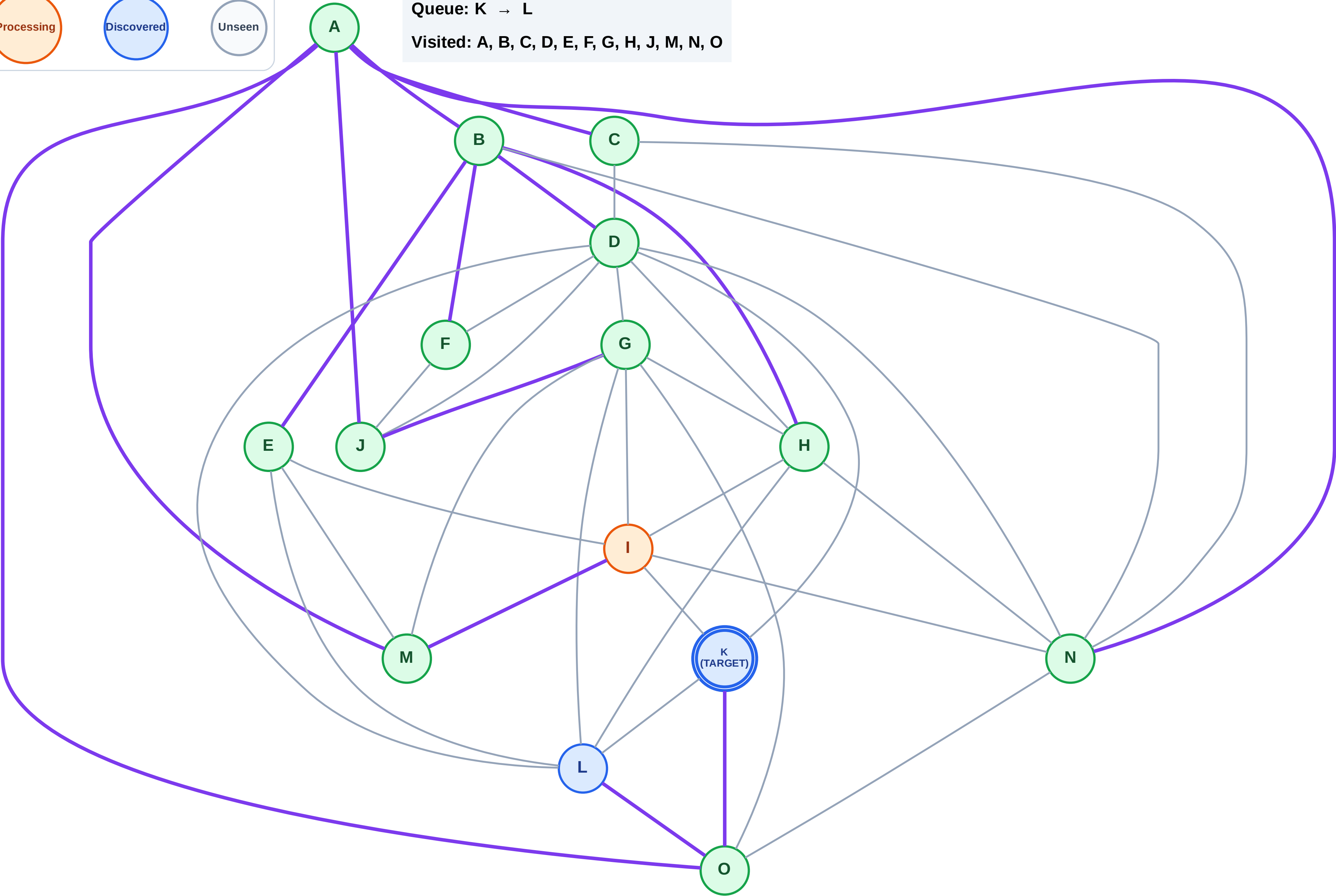
Processing

Discovered

Unseen

Queue: K → L

Visited: A, B, C, D, E, F, G, H, J, M, N, O



BFS Traversal

Step 27: No new neighbors from I

Legend

Complete

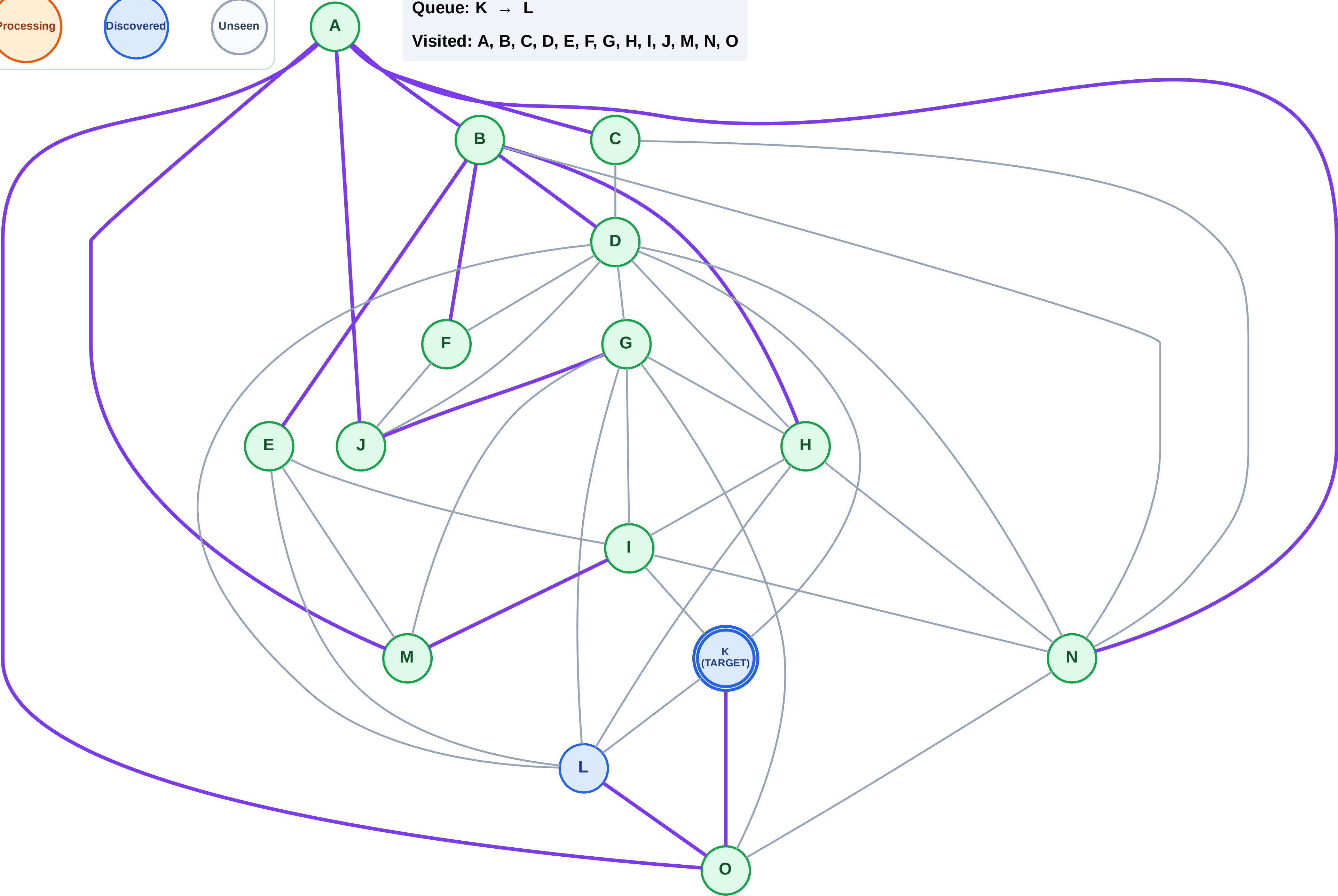
Processing

Discovered

Unseen

Queue: K → L

Visited: A, B, C, D, E, F, G, H, I, J, M, N, O



BFS Traversal

Step 28: Process node K

Legend

Complete

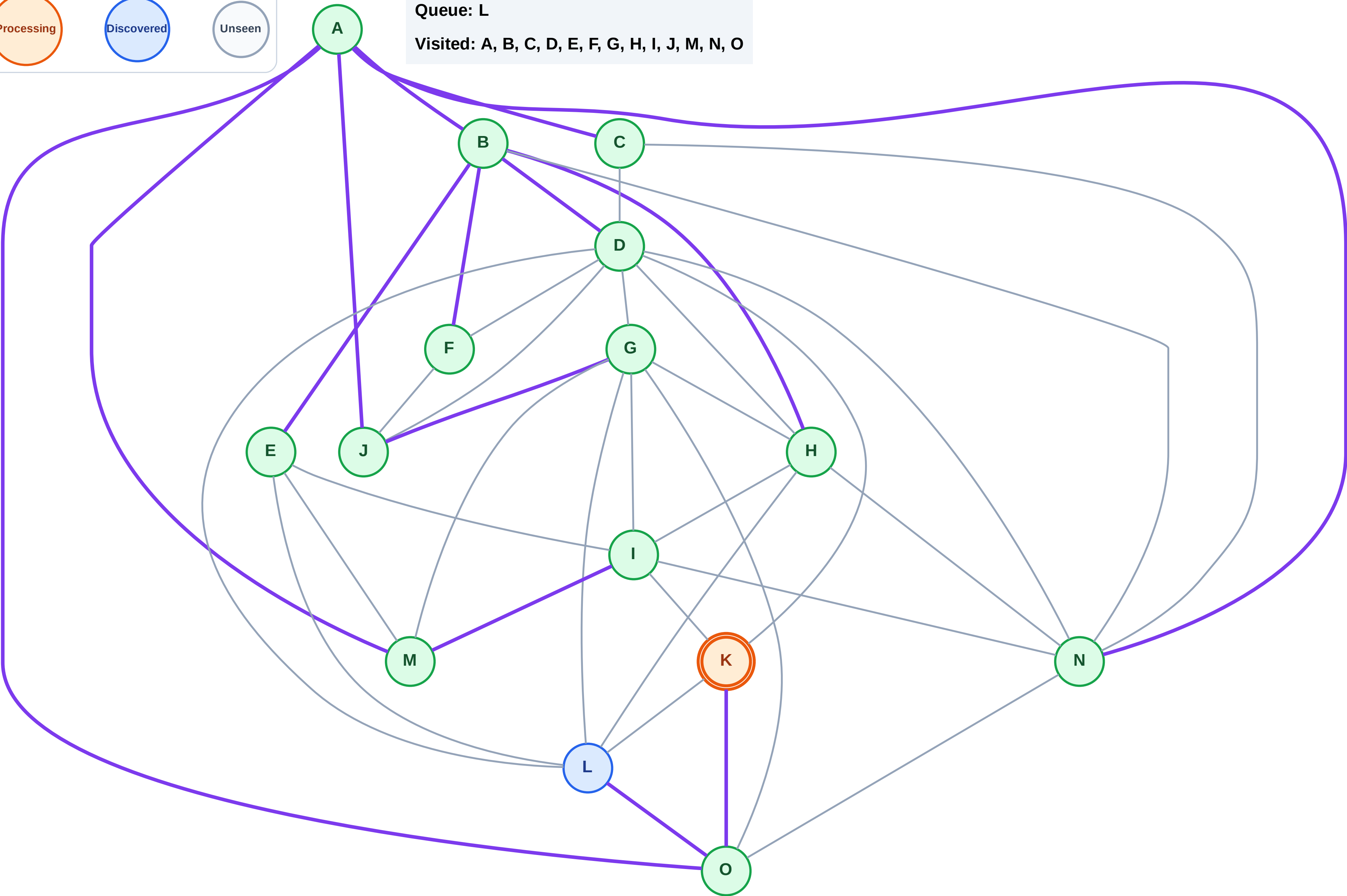
Processing

Discovered

Unseen

Queue: L

Visited: A, B, C, D, E, F, G, H, I, J, M, N, O



BFS Traversal

Step 29: TARGET NODE K REACHED! Path (2 hops): A → O → K

Legend

Complete

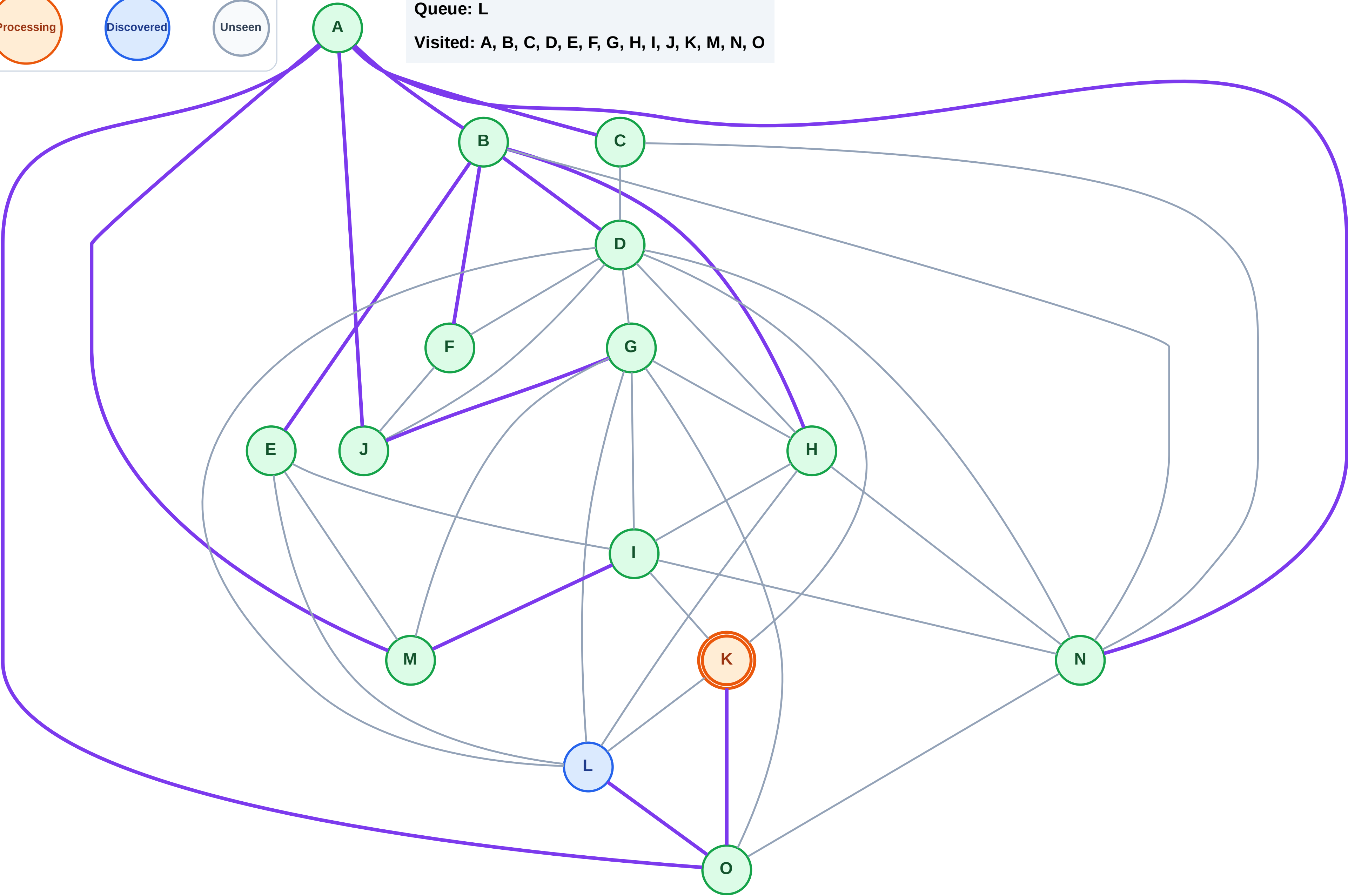
Processing

Discovered

Unseen

Queue: L

Visited: A, B, C, D, E, F, G, H, I, J, K, M, N, O



DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

Complete

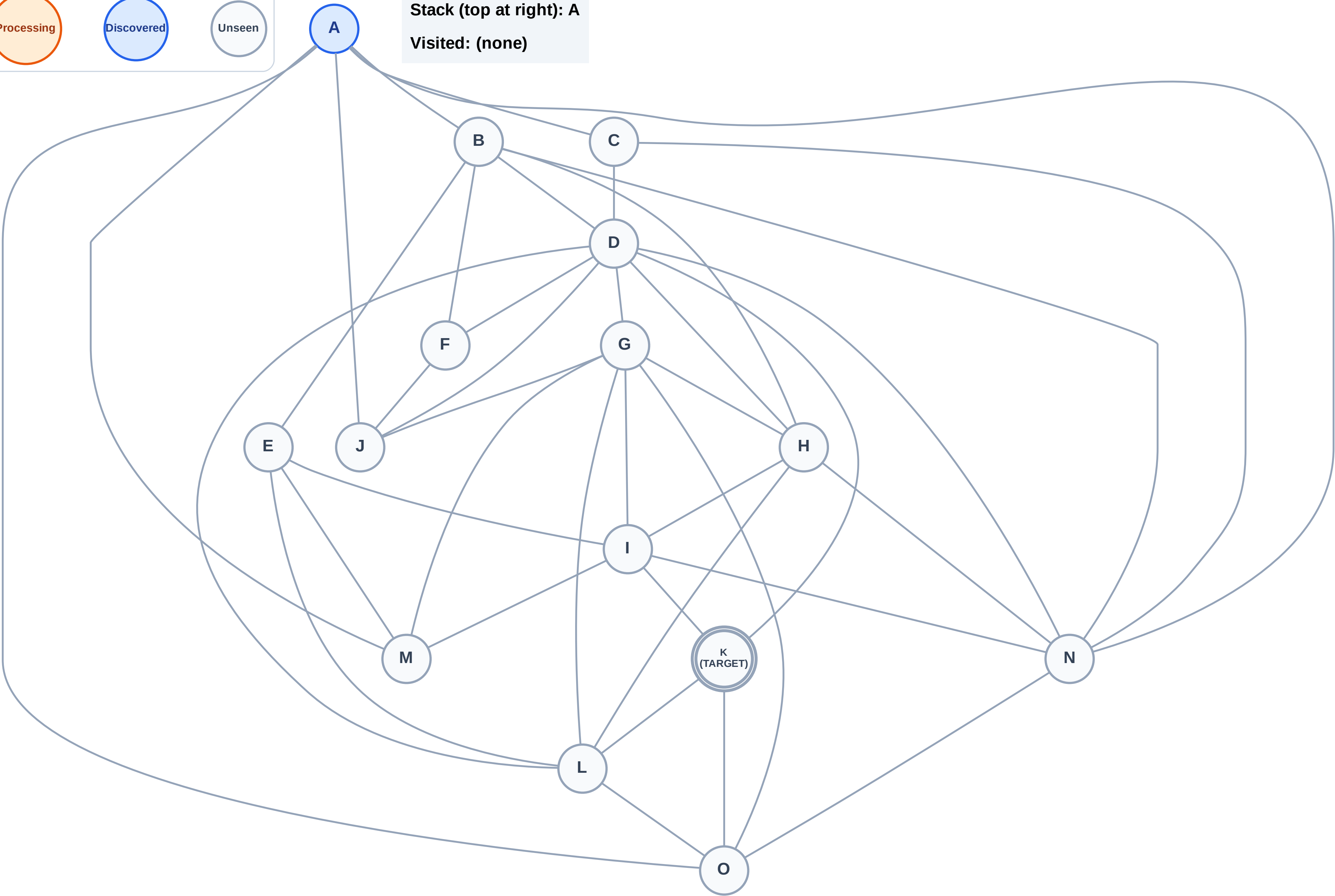
Processing

Discovered

Unseen

Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

Complete

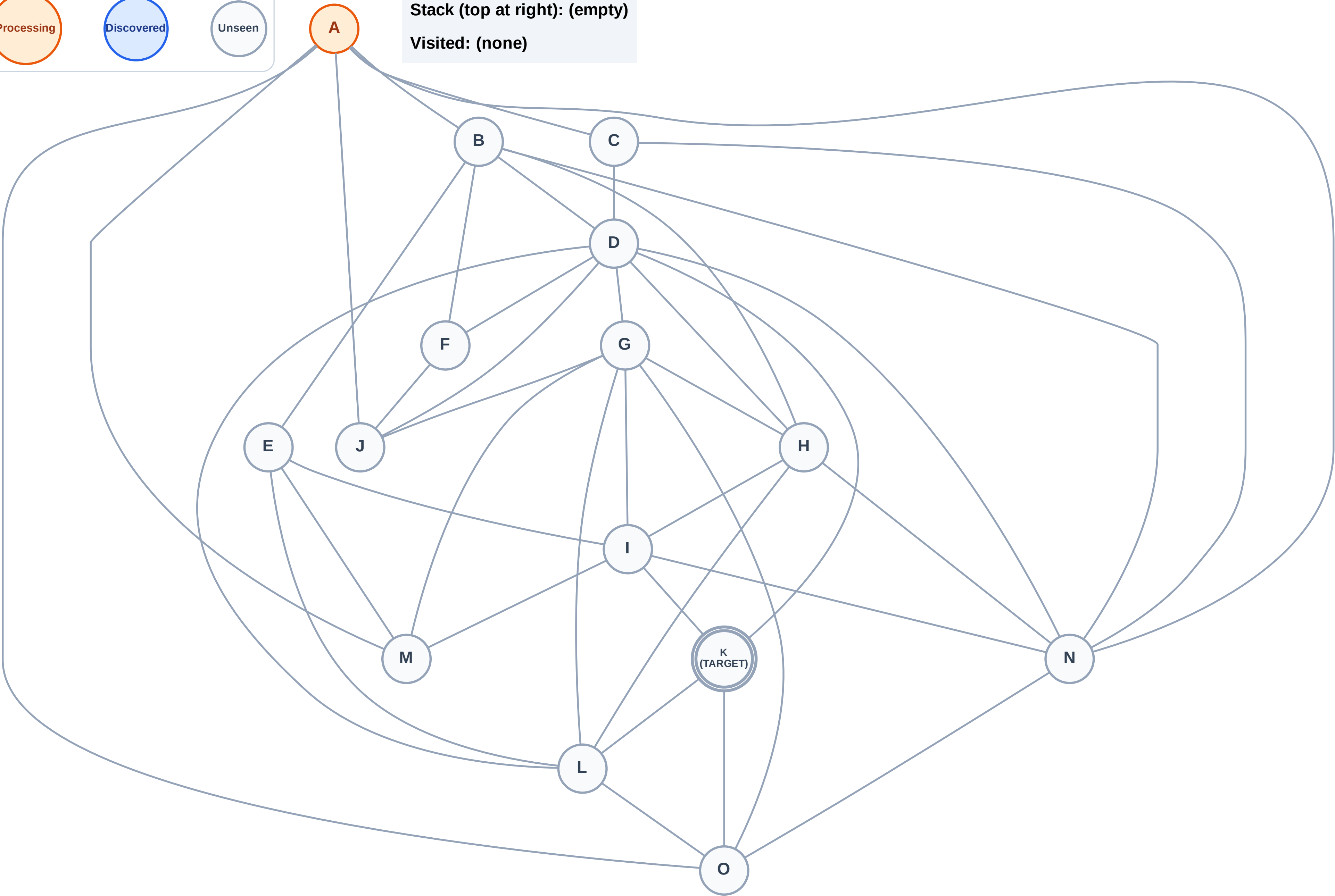
Processing

Discovered

Unseen

Stack (top at right): (empty)

Visited: (none)



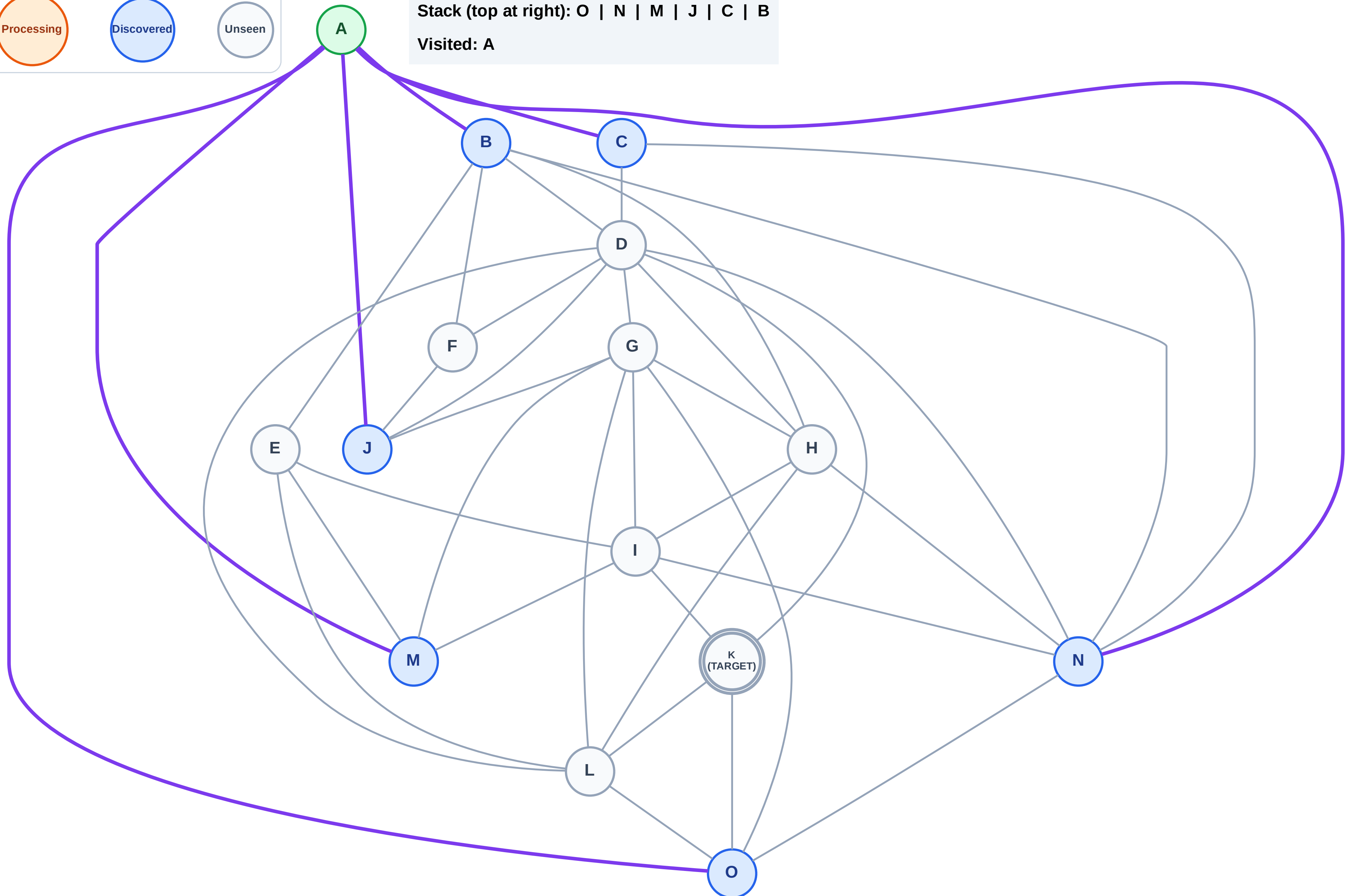
DFS Traversal

Step 3: Discover O, N, M, J, C, B from A

Legend



Stack (top at right): O | N | M | J | C | B
Visited: A



DFS Traversal

Step 4: Process node B

Legend

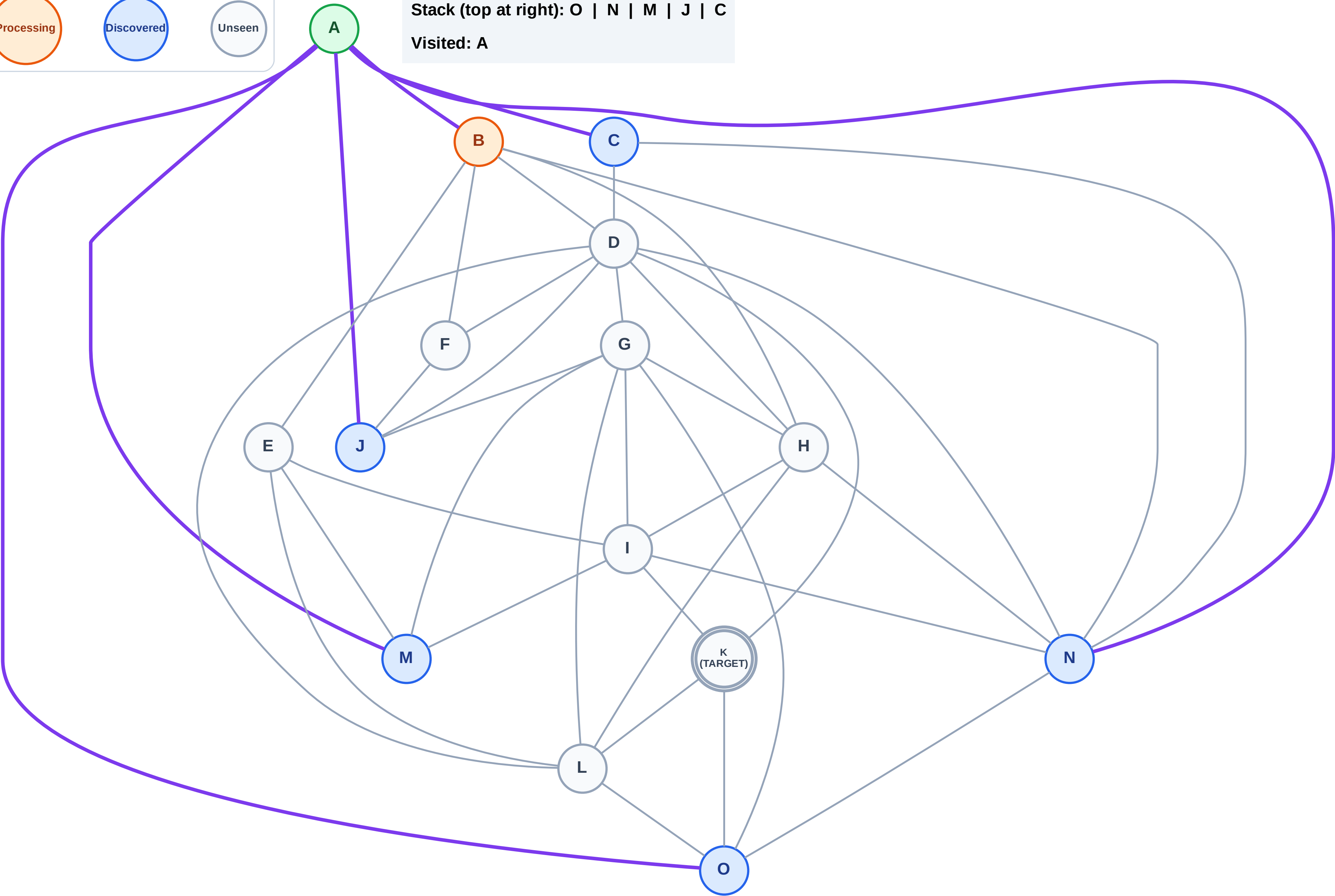
Complete

Processing

Discovered

Unseen

Stack (top at right): O | N | M | J | C
Visited: A



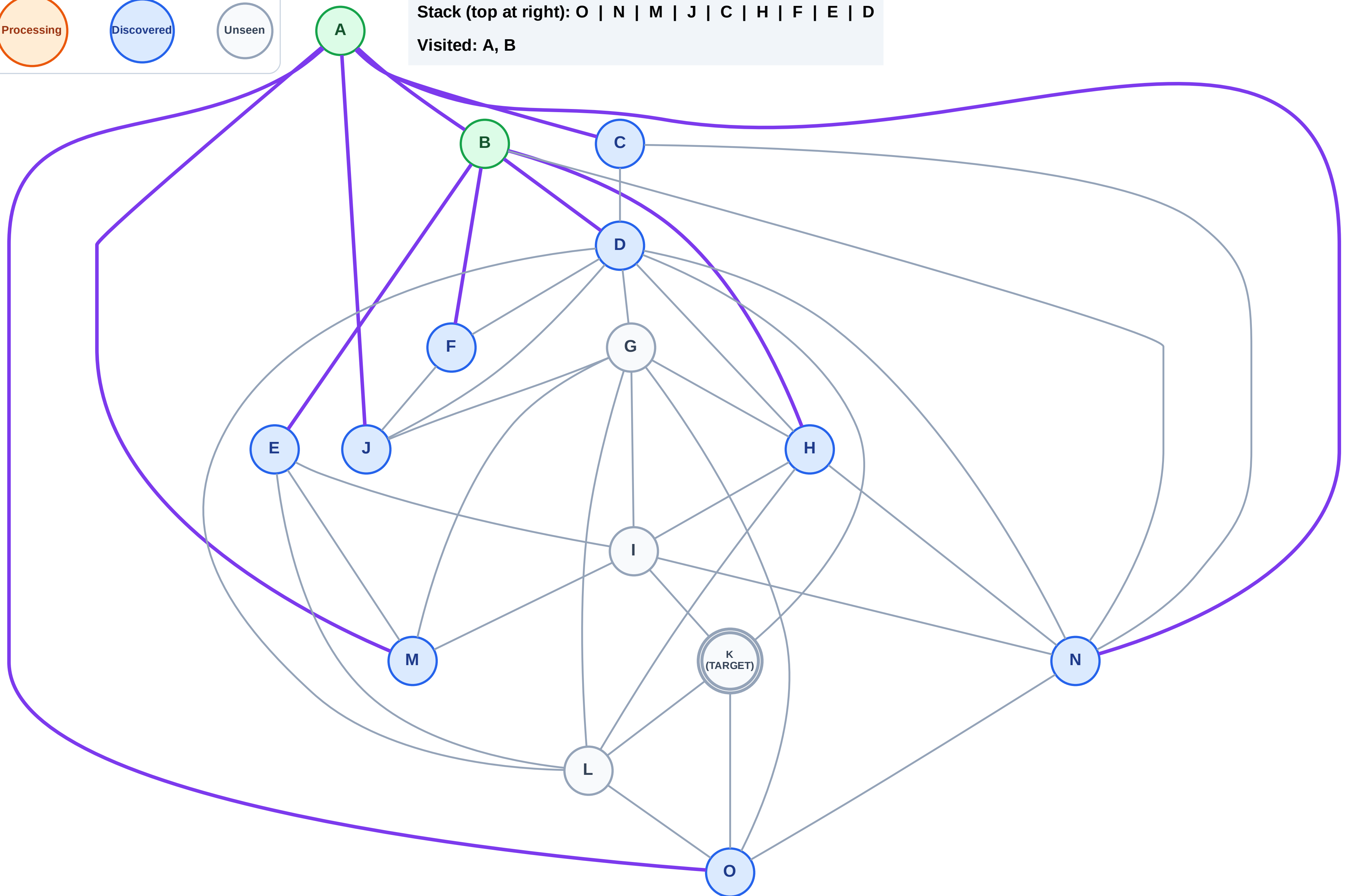
DFS Traversal

Step 5: Discover H, F, E, D from B

Legend



Stack (top at right): O | N | M | J | C | H | F | E | D
Visited: A, B



DFS Traversal

Step 6: Process node D

Legend

Complete

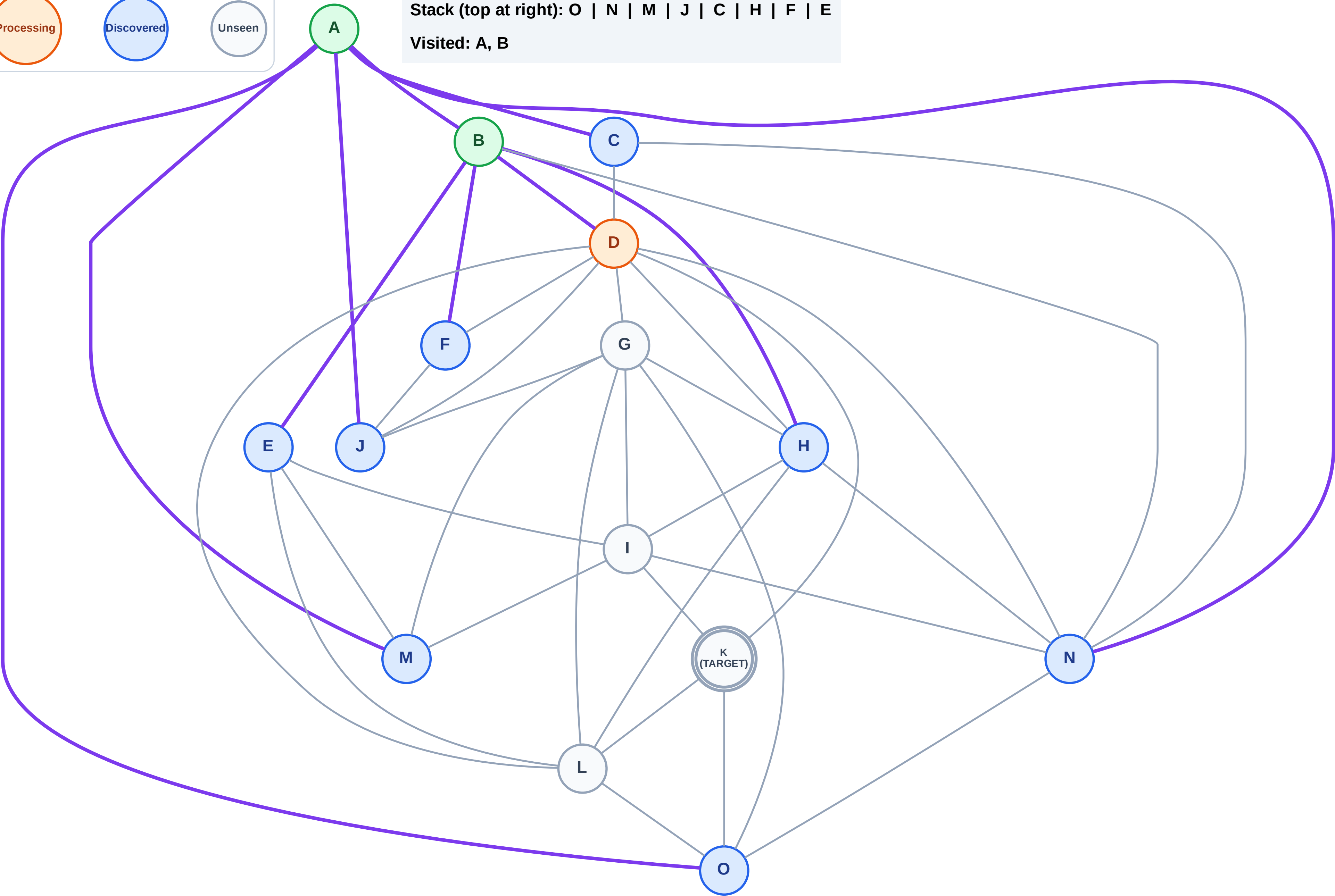
Processing

Discovered

Unseen

Stack (top at right): O | N | M | J | C | H | F | E

Visited: A, B



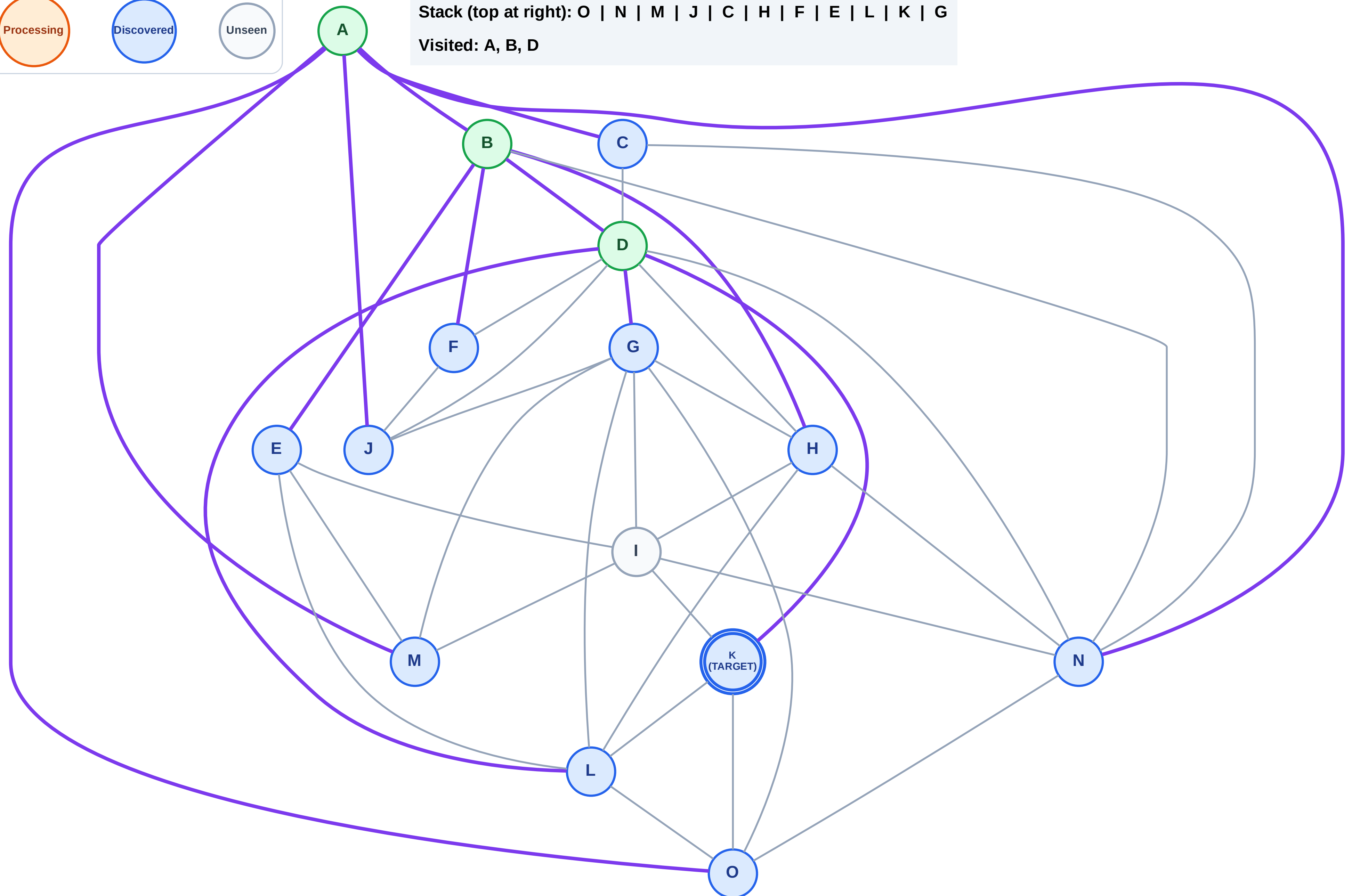
DFS Traversal

Step 7: Discover L, K, G from D

Legend



Stack (top at right): O | N | M | J | C | H | F | E | L | K | G
Visited: A, B, D



DFS Traversal

Step 8: Process node G

Legend

Complete

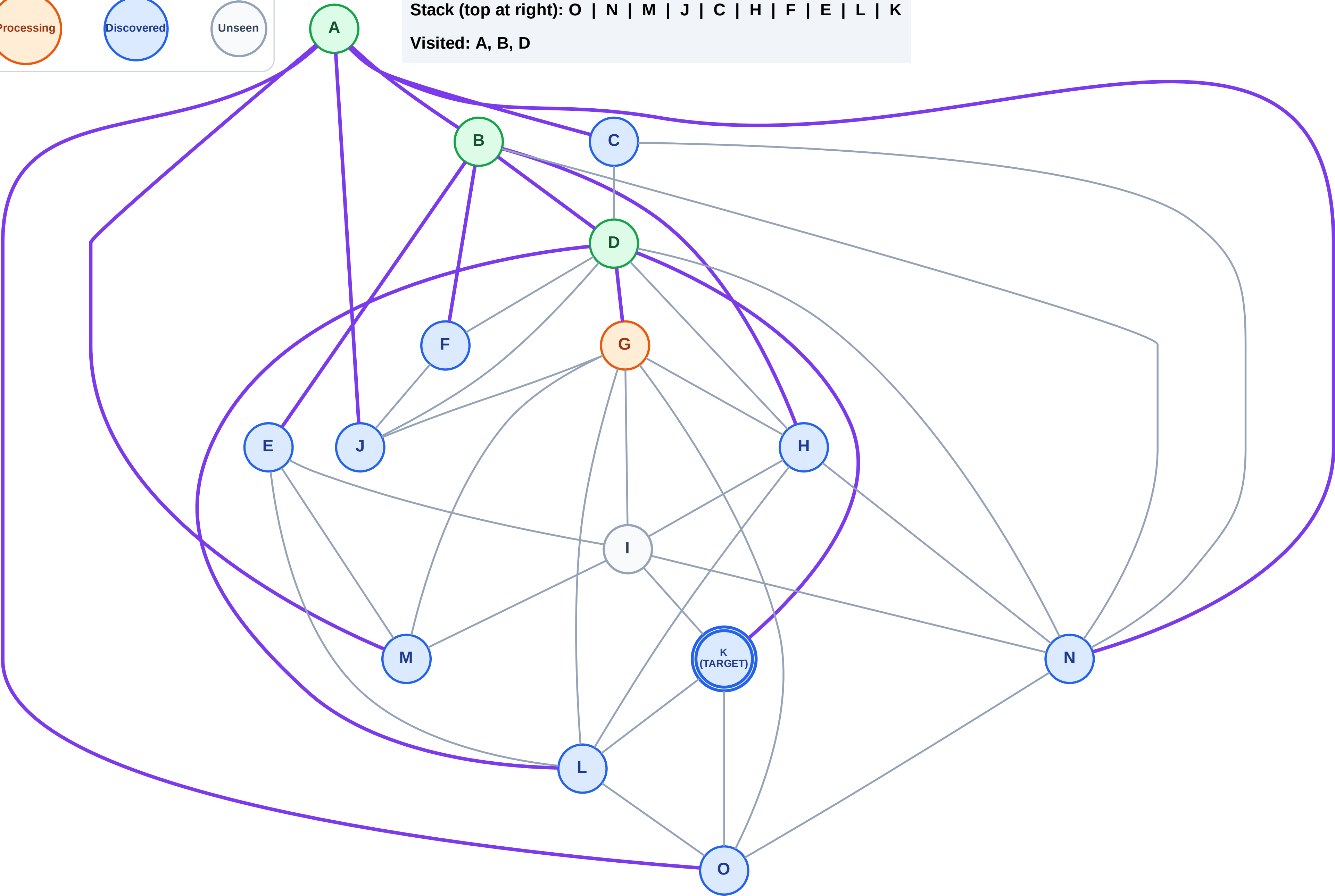
Processing

Discovered

Unseen

Stack (top at right): O | N | M | J | C | H | F | E | L | K

Visited: A, B, D



DFS Traversal

Step 9: Discover I from G

Legend

Complete

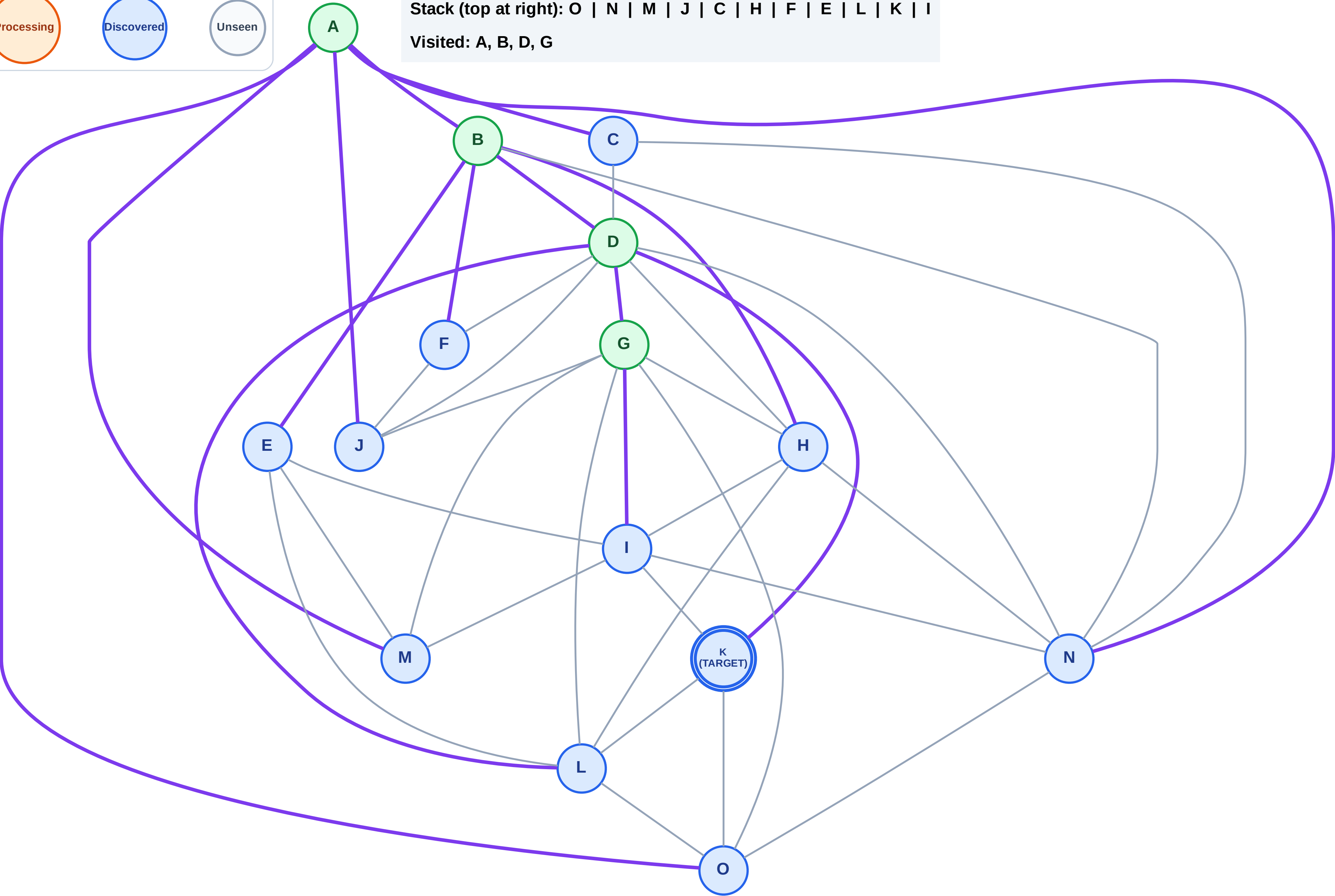
Processing

Discovered

Unseen

Stack (top at right): O | N | M | J | C | H | F | E | L | K | I

Visited: A, B, D, G



DFS Traversal

Step 10: Process node I

Legend

Complete

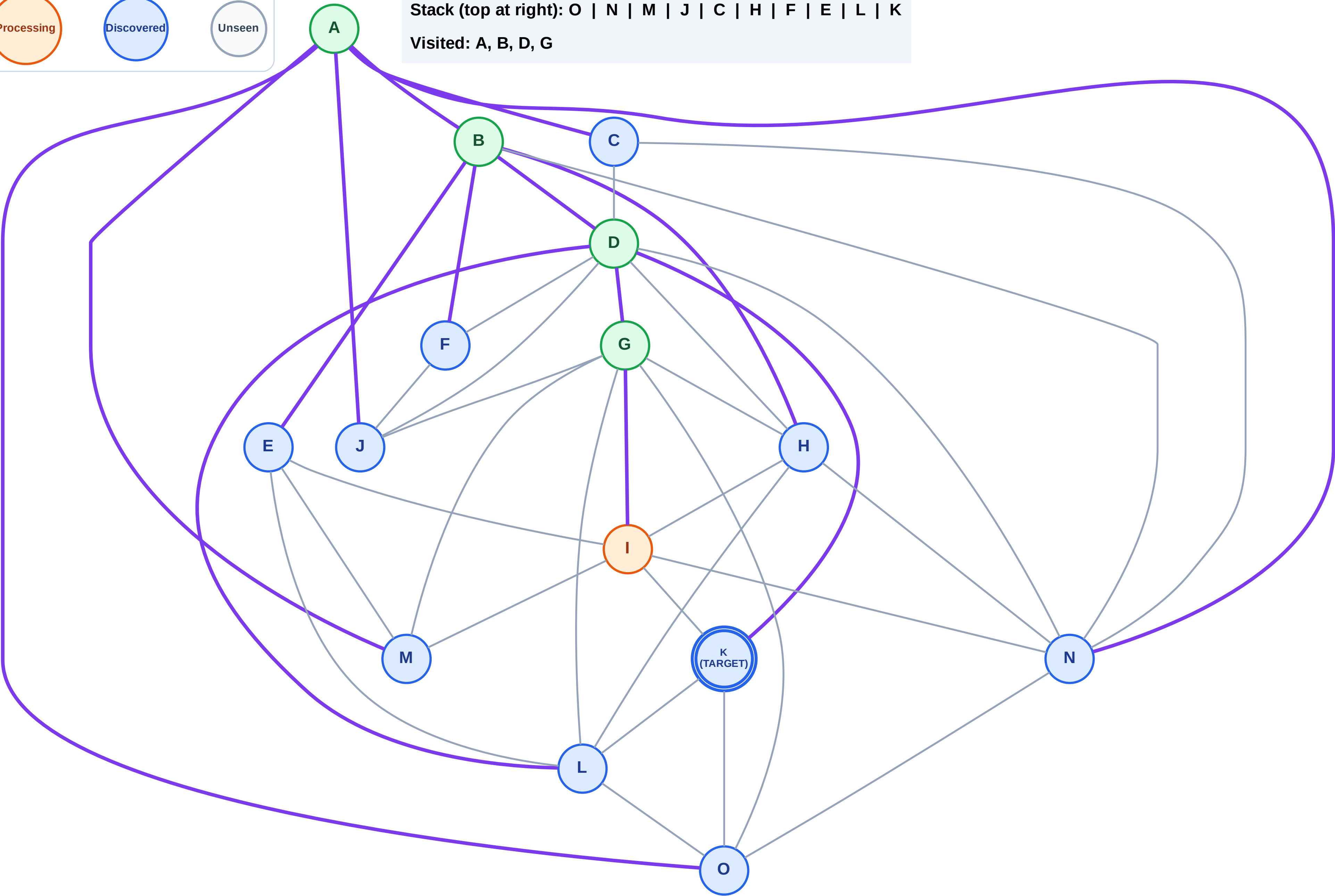
Processing

Discovered

Unseen

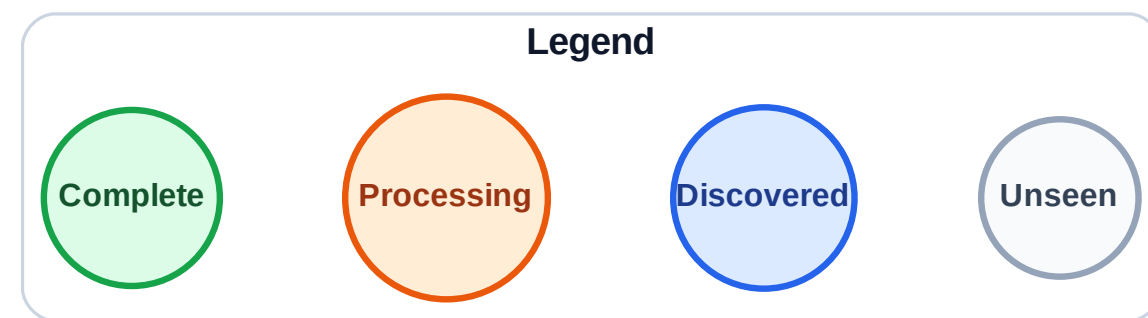
Stack (top at right): O | N | M | J | C | H | F | E | L | K

Visited: A, B, D, G

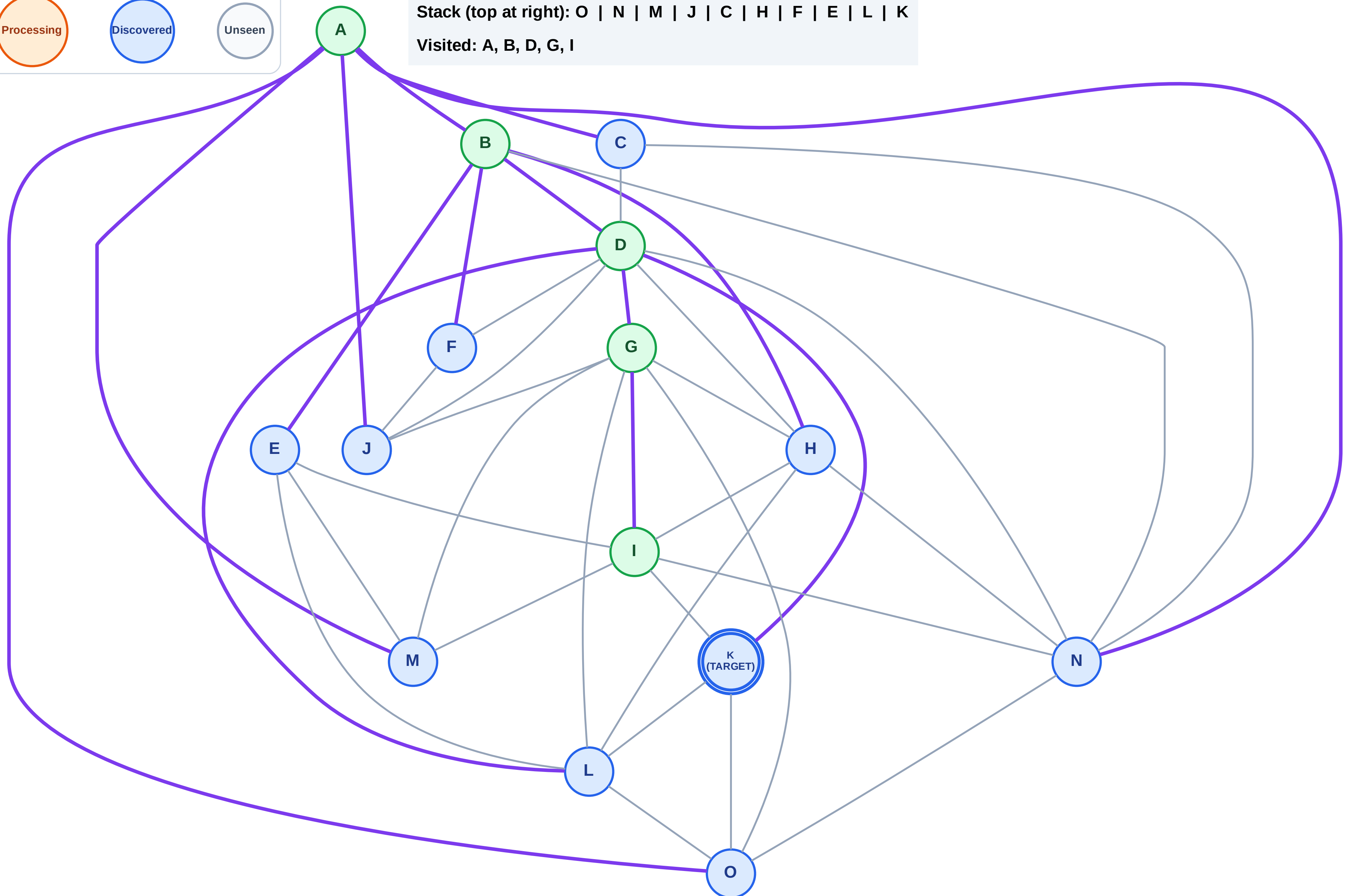


DFS Traversal

Step 11: No new neighbors from I



Stack (top at right): O | N | M | J | C | H | F | E | L | K
Visited: A, B, D, G, I



DFS Traversal

Step 12: Process node K

Legend

Complete

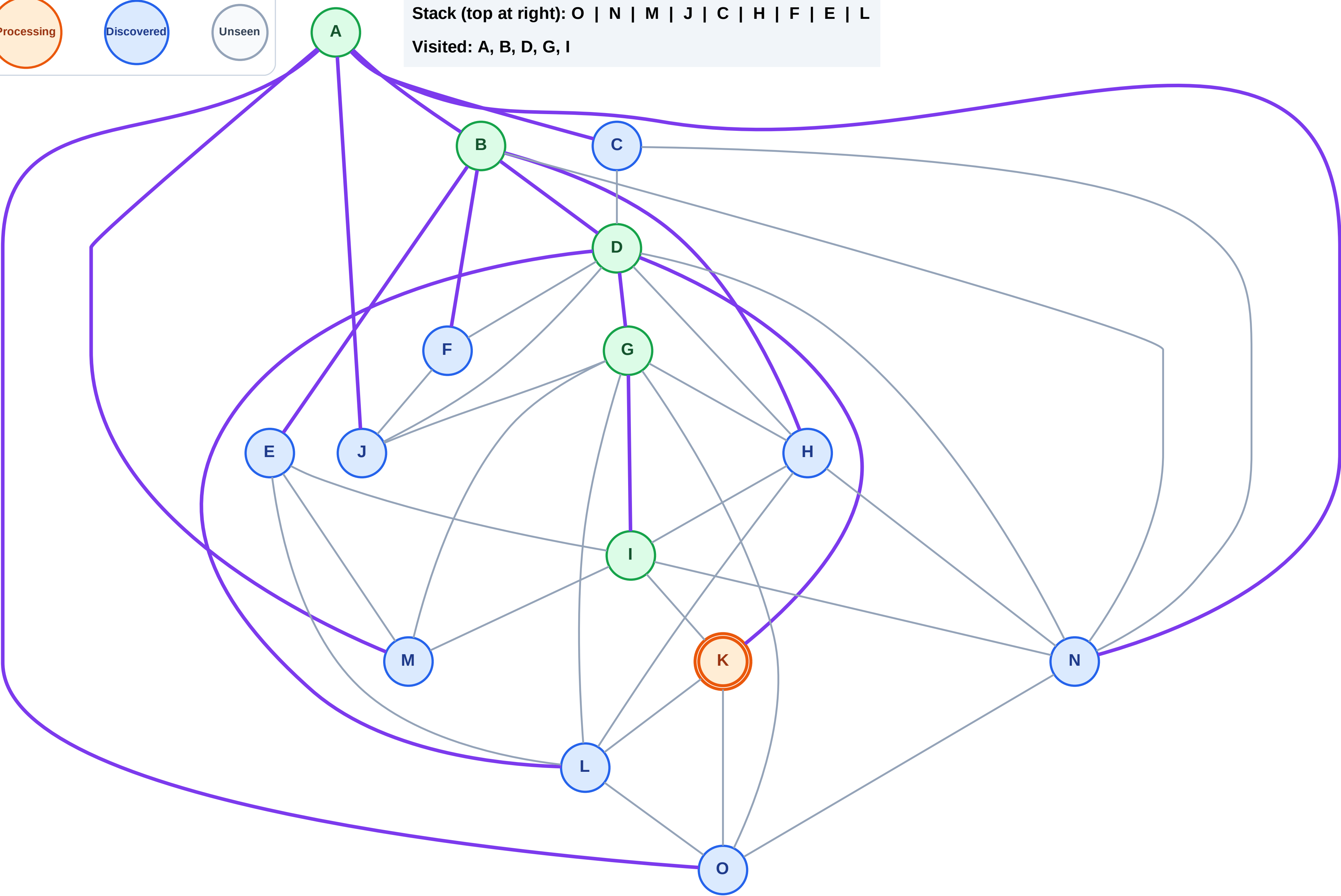
Processing

Discovered

Unseen

Stack (top at right): O | N | M | J | C | H | F | E | L

Visited: A, B, D, G, I



DFS Traversal

Step 13: TARGET NODE K REACHED! Path (3 hops): A → B → D → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): O | N | M | J | C | H | F | E | L

Visited: A, B, D, G, I, K

